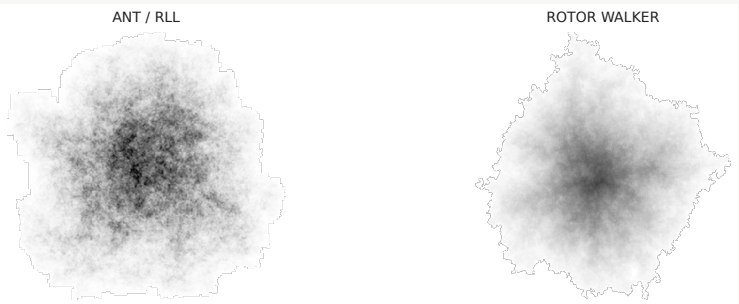
Cube-root growth of most non-highway generalised Langton's ants

Rules of length ≤ 12: calibrated growth and boundary transport

Tim McNamara
Accelerant Ltd, Wellington, New Zealand · tim@accelerant.dev

Abstract. Generalised Langton's ants can spend billions of steps expanding an irregular territory without forming a detected highway. Across 3055 measured rules of length 3–12, most belong to a blob class whose width grows approximately as $t^{\frac{1}{3}}$: doubling the width takes about eight times as many steps. The blob median exponent is 0.333, consistent with $\frac{1}{3}$ at roughly ± 0.01 after calibration against a rotor walker with a proved growth exponent [1]. The census runs to 2^{27} steps, with longer follow-ups on selected rules. Ant and rotor accumulated visit profiles also agree, but surface width grows faster in the examined ant cohorts. A three-factor growth-rate identity locates departures from the cube root in drifting boundary occupation, without deriving its cause. Rule letters predict coarse differences in the probability of entering fresh ground from the edge (held-out $R^2 = 0.81$ for three-letter group means, against 0.66 for a chirality model); an eight-letter prefix lookup predicts $\ln \kappa$, the relative boundary occupation, on fresh stationary rules of lengths 13–14 ($R^2 = 0.633$), including information beyond early area. This restricted prediction leaves the transport mechanism unexplained. Logarithmic growers, faster growers and a rule that switches after billions of steps limit any universal or asymptotic claim.



Same bulk growth. Different measured edge growth.
Occupation fields at 2^{22} steps; darkness shows visits per cell in units of $\frac{t}{A}$, on a shared scale from 0 to 4 (values above 4 clipped).
Each territory is fitted to its panel.

Key results

- **Doubling the width takes about eight times the steps.** The 2630 blob rules have median width exponent 0.333 rotor calibration supports $\frac{1}{3}$ to about ± 0.01, rather than to three decimal places (Section 4).
- **Similar interiors hide different edges.** Ant and rotor accumulated visit profiles agree within 5% in the compared radial shells. Yet surface-width growth exponents are about 0.22–0.25 for the examined ants and 0.10–0.13 for the rotor. The profile is a conditional consistency check; the edge gap is measured at matched step counts, not matched sizes (Section 7).
- **Departures from the cube root appear in boundary occupation.** The 425 logarithmic growers visit the edge increasingly rarely; the 109 fast blob rules typically have rising relative edge occupation. The growth-rate identity locates these deviations in κ, but does not explain why it drifts (Section 5).
- **Rule letters predict two different parts of transport.** For edge-entry probability c, three-letter group means outperform the chirality model on the same held-out rules ($R^2 = 0.81$ against 0.66 Section 6). An eight-letter prefix lookup predicts $\ln \kappa$ on fresh longer rules that simulation has shown to be stationary ($R^2 = 0.633$; the gain over an area model does not survive across all retained rules, and the lookup does not predict stationarity itself; Section 5.3). Neither score establishes the physical origin of occupation.
- **Early growth budgets predict later growth.** A constant budget estimated from early area history reduces typical future-window error by 68.5% against an other-rule baseline. Rare near-stalls produce very large errors, and escaped or incomplete runs are excluded (Section 5.4).
- **The tested memory models do not reproduce the joint regime.** Independent turns, site cookies and edge reinforcement fail to reproduce both ant-like growth and boundary occupation in the tested settings. These controls constrain particular mechanisms; they do not exclude every stochastic explanation (Section 5.5).
- **Billions of blob-like steps need not be permanent.** RLRLLL changes into a column sweeper after about 12.7 billion steps in one blank-board trajectory. Its trigger is unknown, and a board escape is not by itself a highway detection (Section 7.6).

1 The cube-root growth question

Langton's ant [2] is a two-colour automaton whose behaviour is striking: after a chaotic transient of about 10 000 steps it builds a periodic "highway" and escapes. Generalising the turn rule to a string over {L, R} of length n [3] gives a family whose long-time behaviour is, for most rules, unknown. For any rule containing both letters the track is known to be unbounded [3], [4], and the dynamics is computationally hard in general [5], but these results do not determine its growth rate.

How fast does a territory expand when no highway is detected? Our census finds a common answer across thousands of multicolour rules: approximately the cube root of time in width, or $t^{\frac{2}{3}}$ in area. This is slow growth. Under that scaling, eight times as many steps are needed to double the width. The more revealing comparison is with the rotor walker: the bulk growth and accumulated visit profiles agree, while the measured edge dynamics differ. The paper asks which transport measurements account for that agreement and where the explanation stops.

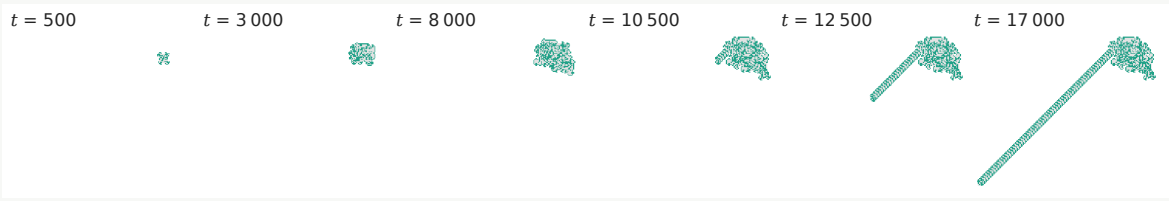


Figure 1: Langton's ant (RL) at six moments, all drawn at the same scale. For about 10 000 steps it wanders and builds an apparently random patch; then it falls into a repeating 104-step cycle that moves it two cells diagonally each time [2] (our sweep reproduces period 104 and onset 9976 for RL: Audit 1), laying a straight "highway" that continues indefinitely. The rules studied here formed no detected highway within the census horizon.

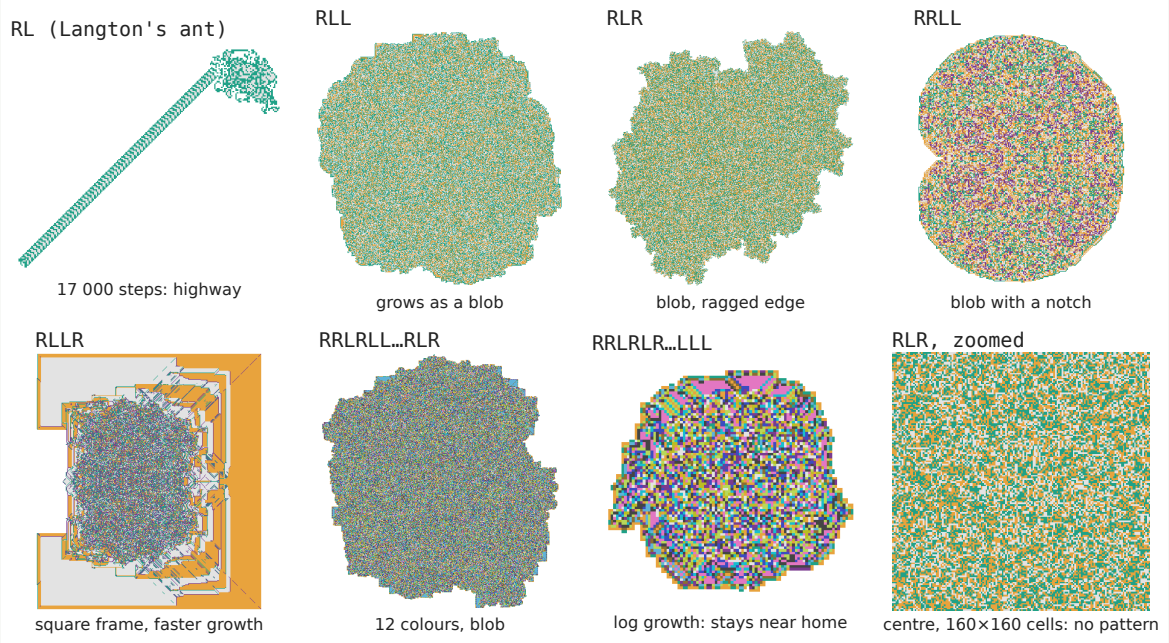


Figure 2: What the ants draw. Each colour is one of the n states a cell can be in; blank is never visited. Top left: classic Langton's ant (RL) after 17 000 steps, a tangle followed by a straight "highway" that runs off forever. The other panels show rules with no detected highway within the census horizon, after 2^{25} (33 million) steps: most grow into a roughly round blob whose texture looks random up close (bottom right), but the shapes differ (notched, ragged, or, for RLLR, a square frame) and one (RRLRLR…LLL) barely grows at all, staying near its starting point. Rule letters give the turn on each colour.

The rotor walker supplies a calibrated comparison. Every site holds an arrow that turns by 90° each visit, and the walker follows it. Rotor radius $\sim t^{\frac{1}{3}}$ was conjectured by Priezzhev et al. [6] and supported numerically by Kapri and Dhar [7]. Bou-Rabee and Peres [1] prove that, for independent uniform initial arrows and clockwise square-lattice exits, the range divided by $t^{\frac{2}{3}}$ converges almost surely to a positive finite deterministic constant, with a deterministic convex limit shape. The checked record lists arXiv v1 (24 August 2026) and no journal reference as of 15 October 2026; this does not determine refereeing status. The ant is not an Eulerian walker, so that theorem supplies no ant growth law.

Boundary occupation is the unresolved transport quantity. We measure how often the ant reaches its edge, relative to uniform occupation of the territory, and call this ratio κ. Its value differs moderately reproducibly between rules. Simple string features predict little of that variation, but a longer-prefix lookup predicts it in a restricted stationary cohort. The distinction matters: predicting occupation is possible in that cohort, while explaining the transport that produces it remains open.

Growth in related two-state media predates this study. Flipping rotators implement the two-state alternating-turn rule; earlier work examined expanding randomized regions [8] and subdiffusive displacement and visited-area statistics [9] in random and spatially periodic media, rather than our uniform multicolour starts. Meng and Cohen measured $A \sim t^{0.75}$ and boundary cardinality $B \sim t^{0.50}$ for three periodic 3×3 initial media without observed propagation through 10^9 steps (least-squares fits after about 5000 steps, their Eq. 1) [8]. Their boundary-return and pocket-growth argument directly precedes our boundary-access analysis; neither their medium nor their measured area exponent equals ours.

These comparisons require distinct observables. Mishra et al.'s ensemble displacement, origin-time averages and visited area are different quantities; a displacement-squared exponent of $\frac{2}{3}$ does not establish a territory-area exponent of $\frac{2}{3}$. Likewise, planar rotor widths [7] and finite-size cylinder studies [10] motivate spatial correlation and clock comparisons, but our trajectory slopes alone identify no interface universality class. Our contribution is the calibrated multicolour growth census and its transport comparisons, with explicit finite simulation horizons, rather than a first measurement of territory growth.

Contributions.

1. A census of 3512 rules with growth exponents for 3055 of them, reporting the whole distribution, its dependence on rule length, and the rules that fall outside it.
2. A calibration: the same pipeline applied to the rotor walker (known exponent) and a random walk (negative control), giving the estimator's bias and between-seed noise.
3. An exact accounting of the area growth rate by edge entry, occupation and geometry, with measured factor levels and drift, building on the earlier boundary-access growth picture [8]. This supplies diagnostics, not a dynamical closure.
4. A future-window forecast from early transport measurements, tested across replicated starts and against rotor and random-walk controls.
5. Held-out predictions of edge-entry probability and, within a selected stationary cohort, occupation; tests of interior colour and short-lag transport approximations.

Section 5 and Section 8 separate what is an identity from what is a test, and list what we could not explain. A hypothesis about "homes" that tether some rules to their origin was investigated and is mentioned only in Section 9; we make no claim about it.

2 Rules, measurements and growth-rate decomposition

2.1 Rule census and finite-horizon classification

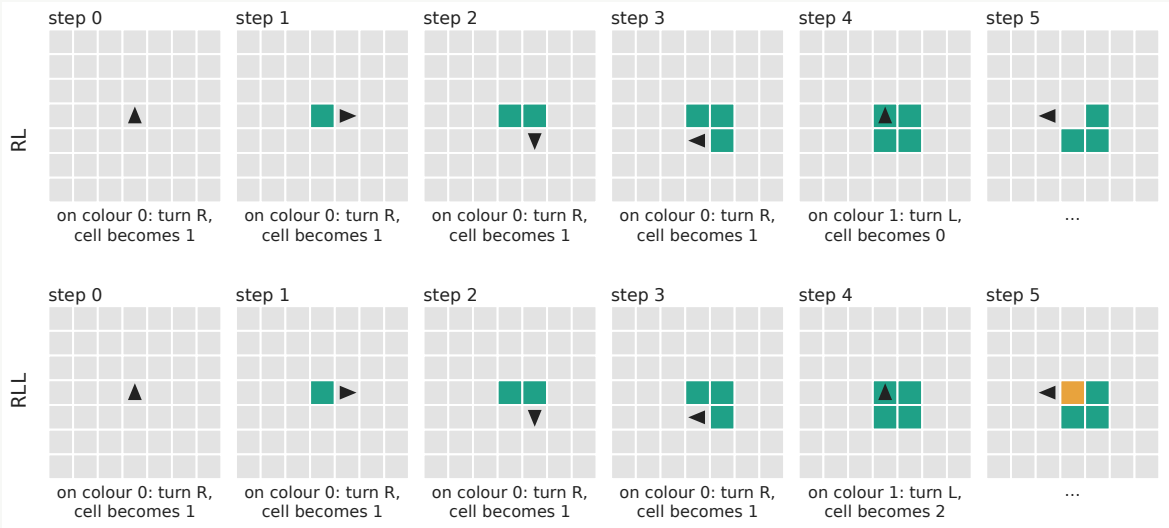


Figure 3: The first steps of two ants. The ant (black triangle) looks at the colour of the cell it stands on, turns left or right as the rule's letter for that colour says, changes the cell to the next colour (cyclically), and steps forward. RL has two colours, so a cell flips between blank and green; RLL has three (blank, green, orange).

A rule is a string $r \in \{L, R\}^n$. The ant has a position, a heading and a grid of colours in $\mathbb{Z}_n$, all initially 0. At each step it reads the colour c under it, turns left or right by `r[c]`, increments the colour to $(c + 1) \bmod n$, and moves one cell. Swapping every L and R gives the mirror-image ant, so we keep only rules starting with R, and drop the trivial rules in which every colour turns the same way.

A sweep of all such rules of length 2–12 (4083 rules) ran each on a blank 8192^2 grid for up to 10^8 steps with a highway detector (Audit 1); 578 escaped or were detected as highway builders (including classic RL). The census studied here is the remaining 3512 rules of length 3–12 (3505 that ran the full budget plus 7 that escaped late).

Each census rule was then run to 2^{27} steps (Audit 2, Audit 1). A rule is called a *builder* (304 rules) if it did not complete 2^{27} steps inside the grid or its steps per bounding-box cell at 10^8 steps were below 20; builders are excluded.

Most are not highway builders: the recognised highways were already removed above, only 5 of the 304 have a recorded candidate highway period (Audit 1) (none confirmed), 289 stay in the grid and fill it slowly (about 2–6 steps per box cell, frame-like structures) and 10 left the grid (Audit 3). We did not check that the 289 form one class.

The census therefore partitions every rule of length 2–12 into four empirical phases, defined operationally by the 10^8-step detector, the builder criterion above and the width-exponent cut below.

Phase	Rules	% of census (3512)
Highway / ballistic (detected, removed before the census)	578	n/a
Frame builders	304	9%
Log-growing (localised)	571	16%
Compact blobs	2637	75%

Table 1: Complete partition of the sweep. The highway rules were removed by the sweep; the other three phases partition the 3512 census rules, .

That about three quarters of the non-highway rules fall into a single macroscopic phase is itself a result. The question of the paper is how growth varies within that phase, so the headline statistics below are for the whole blob class, not for a subset selected on its exponent.

A rule is *log-growing* (571 rules) if the fitted width exponent from $t = 2^{21}$ to 2^{27} is below 0.15, and otherwise *blob* (2637 rules). The exponent distribution is bimodal (Hartigan dip 0.047 on 3208 non-builder rules, $p < 0.001$; a two-component Gaussian mixture has means 0.09 and 0.33 with weights 19% and 81% Figure 4). The dip test only rejects unimodality. The cut at 0.15 was set by eye and is *not* in an empty gap: the log-growers' density tails off across 0.10–0.17 (16 rules within 0.005 of the cut) and the trough is at 0.17–0.26, where bins hold at most four rules. The cut was fixed before the sensitivity analysis, which was registered afterwards (experiment `2026-10-15-alpha-gap`), and it matters little: the blob median width exponent is 0.3312, 0.3326, 0.3328 and 0.3329 for cuts at 0.10, 0.15, 0.20 and 0.25 (spread 0.0003 for cuts from 0.15 to 0.25; 0.0017 when 0.10 is included, which exceeds the 0.001 we predicted in advance; a cut at 0.05 gives 0.3293).

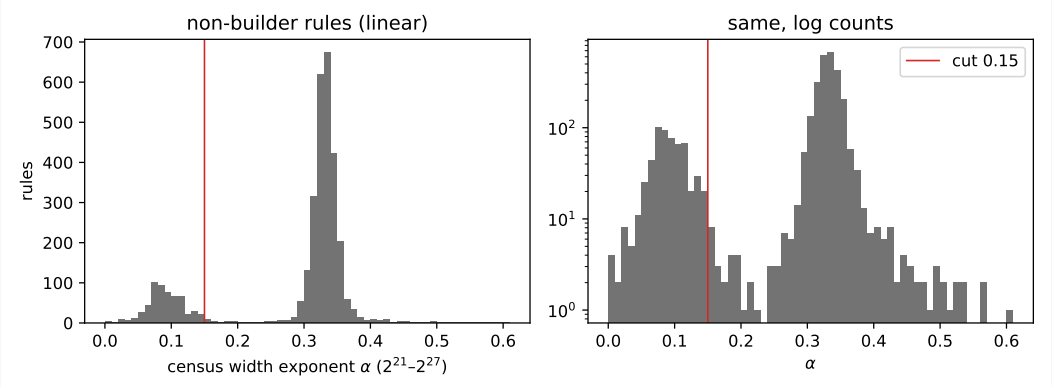


Figure 4: Census width exponent for the non-builder rules; the line marks the log/blob cut at 0.15. There is a density trough, but at 0.17–0.26 rather than at the cut.

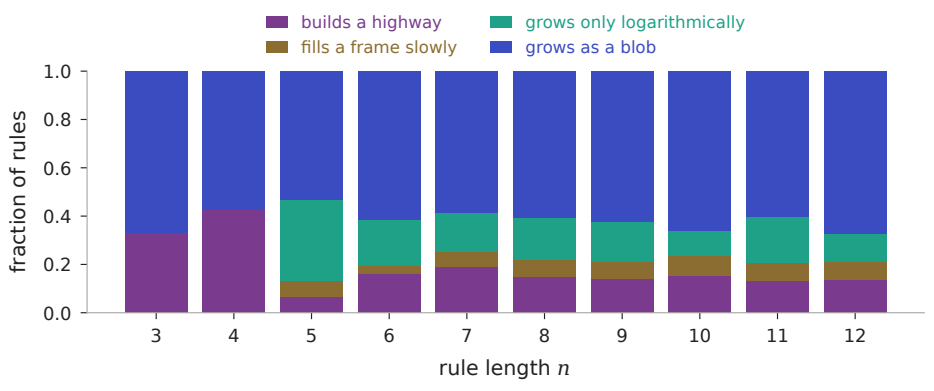


Figure 5: Outcome of every rule in the sweep (4083 rules of length 2–12; the single length-2 rule is RL and is not shown), by rule length. Highway builders are a third to two fifths of the length-3 and length-4 rules and about one in seven from length 6 on. "Frame" builders (mostly rules that fill a frame slowly, a few that leave the grid) are a small slice. The two top classes, blob and log-growing, are the subject of this paper.

2.2 Growth observables, fitting windows and exclusions

The probe (`src/probe.rs`) tracks the *visited* set (every cell the ant has ever stood on), its area A, its *boundary* P (visited cells with at least one unvisited 4-neighbour), and, in octave windows $[2^k, 2^{k+1})$, the number of steps that start on a boundary cell and the number of steps that enter a never-visited cell. It also gives the bounding box, the radius of gyration R_g of the visited set, and the ant's own spread. For each rule we fit log–log slopes over windows $k = 20\ldots26$:

- width exponent α (bounding-box side against t) and R_g exponent α_g;
- area exponent a ($A \sim t^a$; a is always the area exponent and α always the width exponent, with $a \approx 2\alpha$);
- boundary exponent β ($P \sim A^\beta$).

3055 rules were analysed; 153 of the 3208 growing rules had no usable window set (a window contained no new cells) and were not fitted. All analysed rules are kept in the statistics below. 8 of them have a single-window area jump `max_jump` ≥ 4.18 (no unflagged rule exceeds 3.59); they are flagged in the data but not removed, and are all blob-class rules with a median width exponent of 0.42. Dropping them instead would move the blob median from 0.3326 by less than 0.0001.

The unfitted rules are 5% of growing rules, and they are not fast growers: rerun on a 32768^2 grid, none left the grid, all reached 2^{26} steps, and their median area there is 1936 against 316162 for the fitted blob rules, so they are stalled or tethered (Audit 3). If fast growers are missing from the census it is for another reason; the only candidates are the 10 grid-escapers among the builders (0.3% of the census).

The census contains 52 rules that are exact repeats of a shorter rule (u^k, which behave identically to u); we keep them, since removing them moves the blob median width exponent only from 0.3326 to 0.3326.

2.3 Splitting growth into edge entry, occupation and geometry

Growth requires both reaching the edge and stepping beyond it. The probability of entering fresh ground from an edge cell is c. The ratio κ measures how often the walker occupies edge cells, relative to uniform occupation of its territory. A walker can have a high entry probability but grow slowly if it rarely reaches the edge.

Per window of S steps (the window's step count; dA/dt below means $\Delta A/S$ over octave windows) let B be the number of steps that start on a boundary cell and ΔA the number that enter new ground. Boundary membership is evaluated at each step; P and A in the normalization are measured at the window endpoint. For $B > 0$, define

$$c = \frac{\Delta A}{B}, \quad \kappa = \frac{B/S}{P/A}, \quad \frac{dA}{dt} = c \cdot \kappa \cdot \frac{P}{A}. \tag{1}$$

The last equality is an *identity* given these definitions. Measuring its agreement cannot validate a physical mechanism. In particular, κ is an observed occupation ratio, not a closure that predicts how the walker reaches its boundary. Its endpoint normalization also need not equal an average of instantaneous occupation ratios.

Conditionally, if $A \sim t^a$ with $a > 0$, c and κ scale as t^{γ_c} and t^{γ_κ}, and $P \sim A^\beta$, matching powers gives

$$a = \frac{1 + \gamma_c + \gamma_\kappa}{2 - \beta}, \quad \alpha \approx \frac{a}{2}. \tag{2}$$

Equation 2 is also algebra under these scaling assumptions, so agreement between the measured a and Equation 2 (median error 0.008 over blob rules) checks the fits for consistency only. Fixed-ratio octave increments have the same time power as a derivative for a positive power law, but independently fitted slopes need not satisfy this relation exactly; logarithmic growth is outside this power-law derivation. The approximation $\alpha \approx \frac{a}{2}$ additionally requires area to scale as the square of the measured linear size.

The physical hypothesis is stationarity, not factorization. If $\gamma_c = \gamma_\kappa = 0$ and $\beta = \frac{1}{2}$, Equation 2 gives $a = \frac{2}{3}$. These conditions can fail, but measuring them on the same trajectory describes a growth regime rather than explaining why it occurs. Uniform occupation would supply the stronger closure $\kappa = 1$ for a fixed territory; approximate interior colour uniformity does not supply that closure. We do not derive either the stationary regime or its rule-specific occupation from the microscopic rule.

Geometry contributes to the growth prefactor too. In a continuous stationary approximation, writing $P = G\sqrt{A}$ gives $A(t) \approx \left[A(t_0)^{\frac{3}{2}} + \left(\frac{3}{2}\right)c\kappa G(t - t_0)\right]^{\frac{2}{3}}$. Thus a prefactor prediction needs independent values for c, κ and the geometric amplitude G, as well as their persistence. Same-window collapse using measured factors supplies none of these independently; finite octave budgets require the discrete expression in Section 5.4.

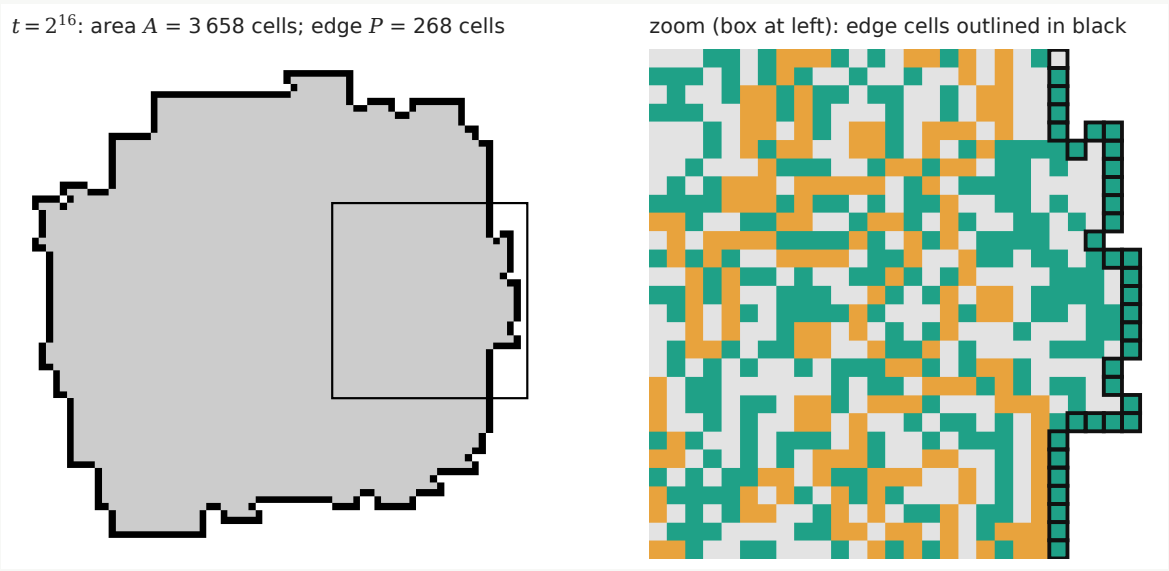


Figure 6: The quantities in Equation 1 on a real territory (RLL, 2^{16} steps). Grey: cells visited so far, area A. Black: the *edge*, visited cells with at least one never-visited neighbour, size P. Only a step taken from an edge cell can enter new ground; c is the fraction of edge steps that do, and κ compares the share of time the ant spends on the edge with the share of the territory that is edge. The right panel zooms on the boxed stretch of coastline.

2.4 Deterministic simulation and reproducibility records

All simulation is deterministic and seeded: rules start from a blank board, so a rule has one trajectory; seeds enter in stochastic walkers and patched-start controls. The fast simulator is tested step for step against a reference implementation. Census summaries, prefix-prediction summaries and forecast errors are read from saved analysis outputs; some follow-up values are transcribed from their recorded summaries. Bootstrap intervals use fixed seeds. Exact producers, recipes and registration records are collected in the appendix audit.

3 Calibrating finite-window bias on the rotor walker

An exponent fitted over seven octaves from 2^{20} to 2^{26} is not the asymptotic exponent. We therefore ran the identical pipeline on 64 rotor-walker seeds (random initial arrows, 2^{27} steps), whose width exponent is $\frac{1}{3}$ and area exponent $\frac{2}{3}$ [1], and on 16 random walks as a negative control.

	width exp. α	R_g exp. α_g	area exp. a	edge exp. β
truth (rotor)	1/3	1/3	2/3	–
rotor, median	0.326	0.339	0.687	0.523
rotor, between-seed sd	0.013	0.006	0.008	0.027
ants (core), median	0.333	0.336	0.679	0.525
ants (core), between-rule sd	0.014	0.011	0.022	0.023
random walk, median	0.498	0.467	0.93	0.93

Table 2: Same pipeline on three systems. The rotor has known exponents; the random walk is a negative control (area exponent near 1, boundary exponent near 1). The "ants (core)" rows are selected on $0.30 < \alpha < 0.37$, which truncates their sd: it understates the between-rule spread and flatters the comparison with the rotor's sd. Figure 9, over all blob rules, is the fair comparison.

Findings.

- (i) The pipeline recovers the rotor: the radius-of-gyration exponent is 0.339 (sd 0.006) against the true 0.333, the width exponent is slightly low (0.326) and noisier, and the area exponent overshoots by 0.02. We therefore treat the finite-window bias of a as $+0.02$ and prefer the R_g estimator, which is the least noisy.
- (ii) The random walk is clearly separated (area exponent 0.93).
- (iii) The ants' R_g exponent 0.336 and area exponent 0.68 sit within the rotor's bias and sd.
- (iv) The boundary exponent of both is about 0.523 (rotor) and 0.525 (ants), not 1/2; it is a finite-window effect shared with the rotor and not evidence of a rough edge.

To check the pipeline further we reproduced the published rotor measurements of Kapri and Dhar [7] from 1000 seeds (radius exponent local slopes 0.263, 0.315, 0.306, 0.346, 0.332, 0.334, tending to $\frac{1}{3}$; boundary-width exponent local slopes falling from 0.19 to 0.132, against their reported value $\alpha/3 \approx 0.13$).

A reservation: for the rotor, the true limit exponent is known, but only by proof for the radius; our statements about its β and c are measurements, not theorems.

4 Census evidence for cube-root growth

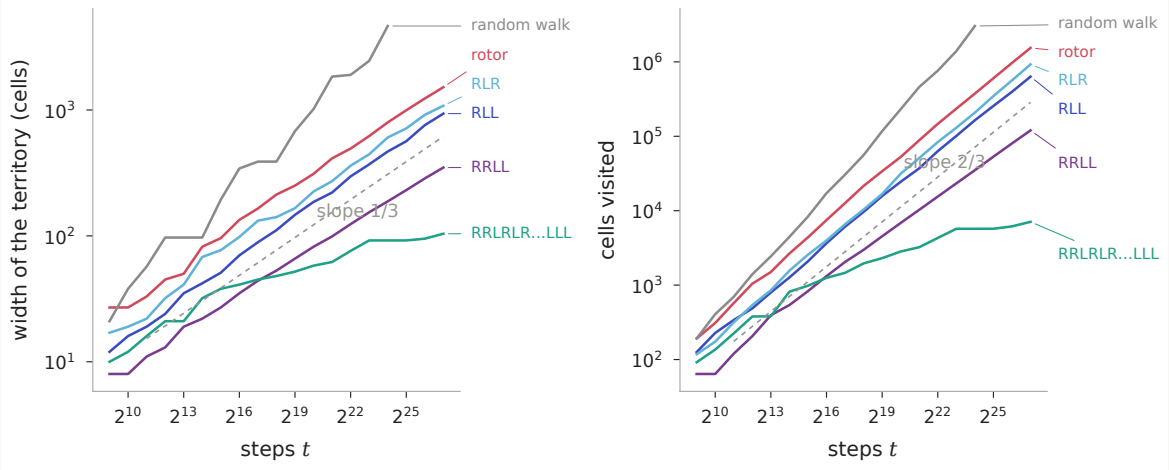


Figure 7: How the territory grows with the number of steps t (log scales, so a power law is a straight line). Three ant rules (blue and purple) and the rotor walker (red) climb parallel to the dashed guide lines, slope $\frac{1}{3}$ for the width and $\frac{2}{3}$ for the visited area; doubling the width takes about eight times as many steps. The log-growing rule RRLRLR...LLL (teal) falls away, and the random walk (grey), the negative control, grows much faster, with width $\sim t^{\frac{1}{2}}$.

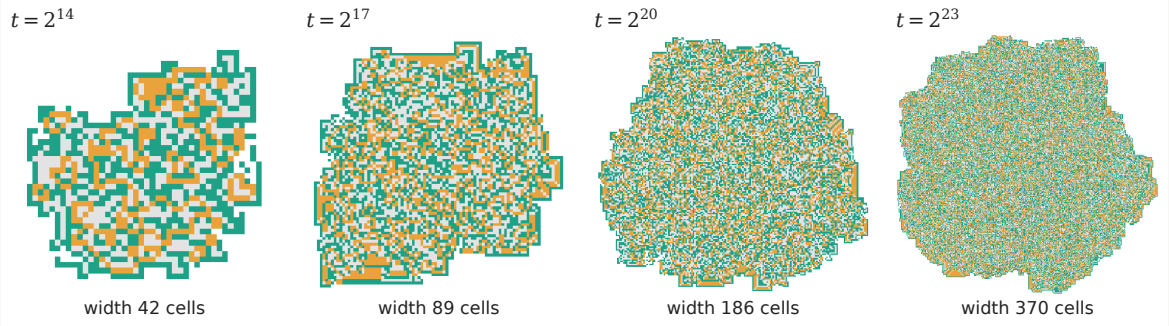


Figure 8: One rule (RLL) at four times, each panel rescaled to the same size. Eight times as many steps buys a territory only twice as wide ($8^{\frac{1}{3}} = 2$), and the shape stays the same round blob: it looks self-similar, and the roundness statistics of Section 7 (anisotropy 0.05–0.07) agree, but we did not test scaling beyond that.

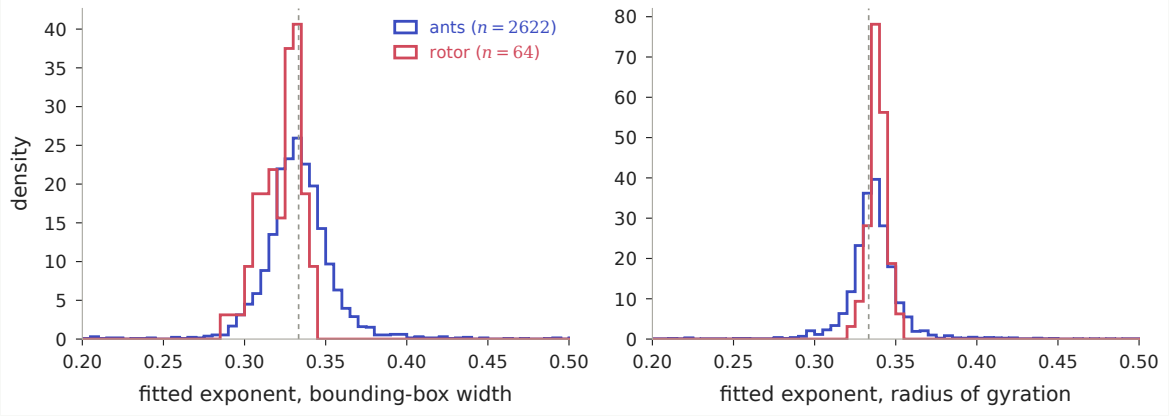


Figure 9: Fitted exponents of 2630 blob-class ant rules and 64 rotor seeds. Dashed line: $\frac{1}{3}$. The ants' distribution is wider than the rotor's: sd ratio 2 (width), 4.3 (R_g).

4.1 Growth exponents and calibration uncertainty

Over the 2630 blob rules the median width exponent is 0.333 (rule bootstrap 95% interval 0.332 to 0.334), the R_g exponent 0.336, and the area exponent 0.679 in the core.

The interquartile range of the width exponent is [0.32, 0.34]. The bootstrap resamples *rules*; it does not include the finite-window bias, which Table 2 puts at about $+0.02$ for a and negative for α.

Against the rotor in the same window the differences (ants minus rotor, medians over all blob rules) are 0.007 for the width exponent, −0.003 for the R_g exponent and −0.008 for the area exponent.

Subtracting the rotor's own bias from the ants' medians gives 0.34 (width), 0.33 (R_g) and 0.658 (area), against $\frac{1}{3}$, $\frac{1}{3}$ and $\frac{2}{3}$. The corrections point in different directions, so the data are consistent with $\frac{1}{3}$ at roughly the ± 0.01 level, not "equal to $\frac{1}{3}$ to three digits"; the correction assumes the ants share the rotor's finite-window bias, which we cannot check.

4.2 Variation with rule length and longer run times

The median is flat in n: Spearman ρ between α and n is −0.018 over blob rules, and per-length medians for $n = 7 \ldots 12$ lie within about 0.01 of one another (table in Audit 6, `by_n`). The spread between rules does not shrink with n in our sample, so we do not claim a rule-length trend or its absence at larger n.

Long runs. Six typical rules were run to 2^{32} steps; their width exponents fitted from 2^{17} are between 0.324 and 0.34, and neither factor in the decomposition drifts (Figure 10).

4.3 Seed variability and reproducible rule differences

The between-rule scatter is mostly run-to-run: re-running 40 rules from 474 random 5×5 starting patches gives within-rule sd 0.016 for α and a between-rule true sd of only 0.004 (intraclass correlation 0.07).

Genuine rule-to-rule differences in α are small; for R_g and a they are detectable (ICC 0.21, 0.33). We cannot say whether they reflect different exponents or different rates of convergence.

Where the rule-specific part of a sits can be partly located (Audit 7, Audit 8 the 99 random rules of the ICC above, window $k = 20$–26). The area is the product of the two box sides and a filling fraction, so $a = \alpha_x + \alpha_y + \gamma_\eta$ exactly, with γ_η the log-slope of $A/(W_x W_y)$. The rule-to-rule variance of a is carried 43% by α_x, 40% by α_y and 17% by γ_η (covariances of rule means, which include run noise); the two sides rise or fall together (correlation 0.6), while γ_η has ICC 0.11, like the sides taken singly (0.09 and 0.12 a itself 0.34). The territories consolidate slowly ($\gamma_\eta = +0.006$, positive for 85% of rules), but rules do not differ in a mainly because they consolidate at different rates. We have not explained why the sides co-vary by rule.

5 Edge transport: predictions and mechanism tests

5.1 Stationary factors and the rotor comparison

Over all 2630 blob rules, the medians (and quartile ranges) of the factors are $c = 0.44$ [0.34, 0.48], $\kappa = 0.91$ [0.73, 1.09] and $\beta = 0.524$ [0.509, 0.539], with territory fill (area over bounding box) 0.76 [0.72, 0.83]. Restricting to the 2405 rules in the band $0.30 < \alpha < 0.37$ (the "core", used in earlier drafts for comparison with the rotor) moves these medians by at most 0.01: c 0.43, κ 0.9, β 0.525.

The drift exponents are small, but the 0.05 threshold is loose: through Equation 2 a drift of 0.05 shifts a by about 0.033 and α by about 0.017, as much as the rule-to-rule differences we are trying to explain. These fractions therefore say that c and κ drift by less than the effect size of interest, not that they are exactly constant, and only the six 2^{32} runs probe much smaller drifts. $|\gamma_c| < 0.05$ for 100% and $|\gamma_\kappa| < 0.05$ for 83% of all 2630 blob rules (the fractions are computed on the whole class, because "core" is itself defined by a width exponent near 1/3 and would make small drifts nearly automatic).

Figure 10 shows the long runs: both factors plateau from 2^{17} to 2^{32} for six rules. These observations characterize the measured cube-root regime; the predictions P1–P4 were written down before the analysis (Audit 9) and specify its proposed stationary factors. P4 predicted β close to $\frac{1}{2}$, and the measured β is 0.525 (core median), equal to the rotor's 0.523 in the same window (Table 2). The original registration's P5 treated reconstruction of the exponent as a mechanism test; that criterion was not independent and is not used as evidence here. Calibration assesses estimator bias, not the validity of a transport closure.

The headline medians are for the whole blob class; the core band is selected on the outcome and appears only as a sensitivity check.

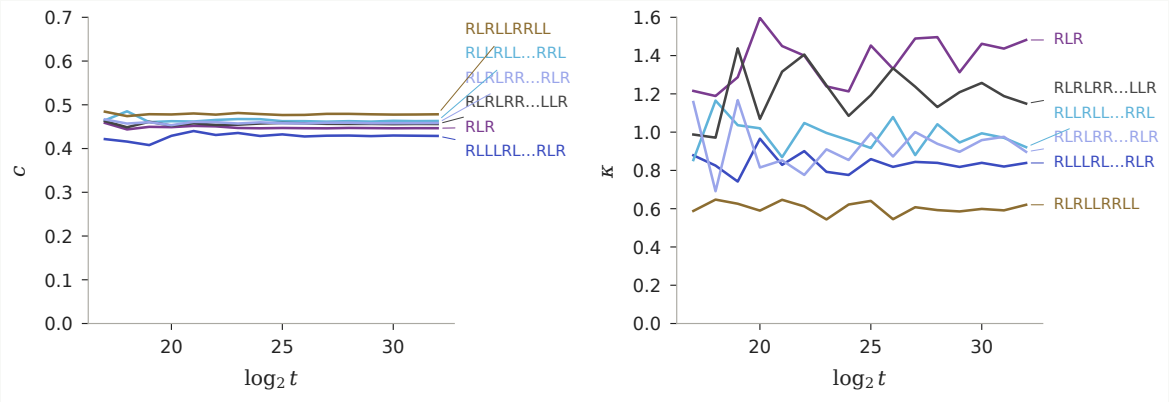


Figure 10: Edge-entry probability c and edge-occupancy ratio κ against $\log_2 t$ for six rules run to 2^{32}. Both are flat after 2^{20}; κ fluctuates by roughly ten percent from octave to octave.

For the rotor the same decomposition gives $c = 0.5$ and $\kappa = 2$; a random walk gives $c = 0.44$. So c alone does not discriminate (the random walk's c is close to the ants'), and the ants' κ (about 0.9) is about half the rotor's.

The cube root does not require the same c and κ as the rotor; it requires that they be constant and that the edge scale as $A^{\frac{1}{2}}$. The ant's edge occupation is *lower* than uniform ($\kappa < 1$ for most rules); we have no explanation of the value of κ.

5.2 Reproducible differences in edge occupation

The relative time spent on the edge varies between rules more reproducibly than the width exponent does. We quantify that reproducibility with intraclass correlations, as in the seed-variability subsection.

Occupation is more rule-specific than the width exponent. Replicated starts show *moderately reproducible* differences (Audit 10 the within-rule sd is nearly as large as the between-rule sd): 99 random rules, each run from 12 random 5×5 patches, have a between-rule sd of κ of 0.44 against a within-rule sd of 0.37 in the core band (ICC 0.58; 0.63 for $\log \kappa$; 58 further core rules give 0.50 and 0.63). The width exponent on this design (Audit 11) has ICC 0.12 (0.065 in the seed-variability subsection, a different set of rules and starts), so κ is much more rule-specific than the measured width exponent.

5.3 Longer prefixes predict occupation in fresh stationary rules

The first eight letters carry information that simple letter summaries miss. The test transfers a prefix lookup from census rules to fresh, longer rules. Its predictive success is restricted to rules selected for a measured stationary regime; it is not a mechanism for κ.

A follow-up registered an eight-letter prefix lookup before sampling 300 primitive rules of lengths 13 and 14, excluding the earlier exploratory sample (W3c; Audit 12). Each rule had eight patched starts, seeds 501–508, run to 2^{27} on an 8192^2 board. Of 2400 runs, 414 escaped; 176 rules retained at least six core-band runs ($0.25 < \alpha < 0.42$, $\kappa > 0$), and 162 also met the stationary criterion $|\text{median } \gamma_\kappa| < 0.05$. Training used stationary census rules of length ≤ 12 only; fresh rules and their outcomes were not added to the lookup.

The lookup predicts mean $\ln \kappa$ from census rules sharing the first eight letters, falling back to shorter prefixes with at least three training rules, or to the training mean. Prefix alone gives strict held-out $R^2 = 0.633$, where the reference prediction is the training mean.

Adding the lookup to a regression on $\ln A(2^{22})$ raises R^2 from 0.497 to 0.737: gain 0.24, rule-bootstrap 95% interval [0.148, 0.347] (2000 resamples, seed 20261045). This passes the registered gain threshold of 0.15 with a positive lower bound. The six-letter gain, 0.117, is below the same threshold, limiting transfer at that prefix length.

The size-controlled model uses measured early area, so its score is not prediction from the rule string alone; the prefix-only score is, conditional on the measured regime filter.

The predictive gain is specific to the stationary cohort. An exploratory crossed comparison (Audit 12, not registered) gives a prefix-8 gain of 0.042 when training on all retained census rules and testing on the same stationary fresh rules, against 0.24 with stationary training. Training and testing on all retained rules gives 0.031 (interval includes zero). Stationary training with all retained test rules gives 0.112, with an interval including zero. The prefix lookup is therefore a **conditional transport predictor**: among rules meeting the finite-window stationarity criterion ($|\gamma_\kappa| < 0.05$), the leading letter sequence carries predictive information about κ ($R^2 = 0.633$). It does not predict whether a rule will be stationary; that requires simulation. This score measures prediction error relative to the training-mean baseline in the selected cohort; it does not identify a causal fraction of occupation variance attributable to the early instructions.

The current conclusion is that a prefix lookup carries predictive information about finite-window κ in the stationary cohort, including information beyond this particular area regression. It is not a derivation of κ, a demonstration of a size-independent effect, or an unrestricted predictor for the census. Core-band and stationarity filters use measured dynamics, the bootstrap holds the training set fixed, and prefix families intentionally overlap between training and test. This tests transfer within prefix families to longer rules, not prediction of unseen families. Why those families share transport properties, and whether the effect survives direct fixed-area sampling and prospective regime selection, remain open.

5.4 Early transport budgets forecast later growth

A same-window identity cannot test persistence. We therefore specified a forecast on existing replicated runs before computing its errors (Audit 17). Write $g = \frac{P}{\sqrt{A}}$; formally Equation 1 gives $\frac{d\left(A^{\frac{3}{2}}\right)}{dt} = \left(\frac{3}{2}\right)c\kappa g$. For an octave ending at t, with duration $S = \frac{t}{2}$, the exact discrete budget is

$$Q = \frac{A^{\frac{3}{2}} - (A - \Delta A)^{\frac{3}{2}}}{S}. \tag{3}$$

The endpoint product $\left(\frac{3}{2}\right)c\kappa g = \left(\frac{3}{2}\right)\left(\Delta\frac{A}{S}\right)\sqrt{A}$ is a right-end approximation, typically 1.112 times the exact budget in this cohort. We forecast with Equation 3. Each trajectory's mean budget over windows ending at 2^{20}, 2^{21} and 2^{22} predicts its increments in $A^{\frac{3}{2}}$ over the next four octaves, ending at 2^{23}–2^{26}, without updating from those outcomes.

A competing model extrapolates a log–log drift fitted to the same three early budgets. The rule-independent baseline is the median early rule-mean budget after excluding the target rule and all its seeds.

There are 1185 complete trajectories across 99 rules, with approximately 12 patched starts per rule; 15 trajectories are excluded by the completeness, nonescape or positive-growth criteria. We apply no outcome exponent filter. For each trajectory, error is the median absolute relative increment error over four forecast windows; the summary is the median of equal-weight rule means.

Early budgets improve the typical forecast. Constant budgets give 8.5% error against 26.9% for the baseline and 23.3% for the drift fit. The reduction is 68.5% (rule-bootstrap 95% interval 61.3–80.6%; 2000 resamples, seed 20261013), passing the stated 30% prediction. Integrating all four forecast increments from the observed area at 2^{22} gives final-area error 5.4% against 17.6% for the baseline.

Calibration separates persistence from a universal forecasting advantage. The same estimator on 64 rotor seeds, with matched windows, gives 6.1% error against 5.7% for the other-seed baseline.

On 16 random-walk seeds (training $k = 17$–19, forecast $k = 20$–23), errors are 60.3% and 58.8%; drift extrapolation still misses by 47.9%. Synthetic $A = Ct^{\frac{2}{3}}$ and $A = Ct$ recover their known constant and square-root budgets respectively.

Near-stalls bound the forecast's reliability despite its typical accuracy. RMS within-rule seed sd of relative increment errors is 188.68 for constant budgets, 918.8 for drift and 8.67 for the baseline; these are ratios, not percentages. One retained trajectory (`RLLRRRLLRLRR@108`) has constant-budget error about 6506 times the observed increment. Near-stalls make relative errors enormous, so the robust improvement does not imply uniformly reliable predictions. Escaped or incomplete trajectories are excluded, and this test cannot forecast their breakouts.

An exploratory check, specified after the primary result, uses the median early budget of the other seeds of the same rule and gives 7.4% error, slightly better than each trajectory's own early budget. This supports persistent rule-level growth budgets in the measured windows; it neither identifies the spatial-memory mechanism nor proves asymptotic scaling. Exact recipes, input hashes and all trajectory errors are recorded in the appendix (Audit 18).

This forecast tests persistence of growth, not the separate transport factors. Equation 3 uses only area endpoints. Its success therefore cannot show that measuring c and κ adds predictive information beyond early area history, or validate their physical interpretation. An independent factor model would have to predict boundary access on held-out trajectories without using their measured boundary occupation or future growth, then improve growth forecasts against an area-history baseline. No such mechanistic closure is established here.

5.5 Mechanism tests: local transport, board memory and boundary trapping

The tests bound what can set boundary occupation by separating three proposed sources: local transport statistics, stochastic spatial memory, and near-boundary geometry. Table 3 collects the resulting constraints. Local summaries leave substantial rule dependence unresolved; the tested forms of stochastic memory do not reproduce growth and occupation together; and similar edge dwell times coexist with widely differing occupation. These constraints apply to the specified models and finite windows. They do not exhaust all stochastic models or establish that deterministic arrow updates are mathematically necessary. A successful mechanism must explain the joint growth and occupation regime, as well as its rule dependence.

Candidate class	Hypothesis	Operational test	Finding	Constraint
1. Local / Markovian	Local turn bias or short strings determine κ.	Held-out linear/ridge fits; edge-colour predictors.	$R^2 = 0.003$ for simple strings; edge predictors $R^2 = -0.05$.	These local summaries do not predict occupation; longer prefixes can.
1. Memoryless transport	Independent persistent turns reproduce the ant regime.	Matched probe; five turn rates, six seeds each.	$a = 0.94$–0.96; $\kappa = 1.3$ –1.45.	Rejects this independent-turn closure of board dynamics.
2. Stochastic board memory	Site cookies or edge reinforcement suffice.	Eight seeds per setting; growth and occupation measured together.	Directional cookies escape; random cookies are diffusive; reinforcement has falling κ.	Memory alone is insufficient in these settings.
2. Arrow randomisation	Noise smoothly interpolates rotor occupation to a memoryless limit.	Six noise levels, eight seeds each; 38 fitted runs.	a rises; κ first rises to 3.6–6.7 while c stays 0.498.	Phase disruption changes transport without changing edge-entry bias.
3. Boundary trapping	Edge dwell or crust depth determines κ.	Layer profiles on 24 rules; 26 matched pairs.	Similar dwell with widely varying κ; depth ordering 13/26.	Local trapping summaries do not determine occupation.
3. Excursion geometry	Return angles and scaled displacement explain κ.	Size controls; held-out forecasts; disjoint-seed matching.	Simple associations are size-confounded; a cutoff association remains.	No established geometric mechanism; cutoff dynamics remain open.
3. Excursion duration budget	A tail of long interior returns sets the boundary share B/S beyond boundary residence and geometry.	Registered 44-rule, 1408-trajectory test; rule-held-out PRESS gain, three area bands.	T1 fails (band 0 gain 0.35, lower bound −0.02); T2 passes (0.12) but persistence is better (0.07).	Not supported at these scales; selected cohort only.

Table 3: Exclusion matrix for candidate mechanisms. Findings constrain the tested closures rather than entire mathematical classes. Designs, windows, uncertainty and qualifications are given below and in the linked audits.

5.5.1 Local features and independent turns leave occupation unexplained

Simple letter features. (Audit 10): a linear regression on string features (turn probability q, n, run count, letters 1–6), fitted on a random half of the 2630 non-bursty blob rules and scored on the other half, has held-out R^2 = 0.003 for κ and 0.002 for its drift γ_κ, against 0.78 for c with the same features and split (a different, simpler model than the chirality model in Section 6, which reaches 0.66 on a larger held-out set); Spearman(κ, q) = 0.08. On 192 rules with at least six core-band runs, leave-one-rule-out R^2 for the rule-mean log κ is -0.06 to -0.04 for ridge on letter-by-position, about 0 for ridge on 2- and 3-letter substring counts, and 0.04 for a prefix-5 neighbour mean. These scores constrain the tested low-order summaries, while leaving nonlinear information in longer prefixes available. **Edge-colour statistics.** (Audit 10): on 40 random core rules the edge-weighted turn probability correlates with c at Spearman -0.84 (a positive control that the instrument can see rule dependence) but with κ at only -0.27, and leave-one-out R^2 of κ on it and the edge-colour distance from uniform is -0.05.

Independent-turn control. The transport closure is tested directly by a memoryless walker that turns right with probability Q and left otherwise, independent of the board, driven through the same probe (Audit 10 $Q \in \{0.25, 0.4, 0.5, 0.6, 0.75\}$, six seeds each, 2^{26} steps on a 32768^2 board; predictions written beforehand in Audit 19, A1d). Its κ is roughly constant at 1.3 to 1.45 (within $\pm 5\%$ across Q from 0.25 to 0.6), but its area exponent is 0.94 to 0.96, not $\frac{2}{3}$.

So independent persistent turns reproduce roughly uniform occupancy but neither the cube root nor the ants' κ (about 0.9) nor the rotor's (about 2.0). This rejects the independent-turn mechanism; it motivates testing board memory without establishing that memory alone is sufficient, or excluding every memoryless mechanism.

5.5.2 Tested stochastic memories do not reproduce the joint ant regime

Spatial memory is a candidate ingredient, but its presence alone need not preserve the ant regime. A once-excited walk follows a directional cookie on its first departure and moves uniformly on revisits [11]; random-arrow cookies remove the ensemble drift. A linearly edge-reinforced walk chooses neighbours with weights initially 1 and incremented by I per crossing [12]. These isolate one-use site memory and persistent edge memory.

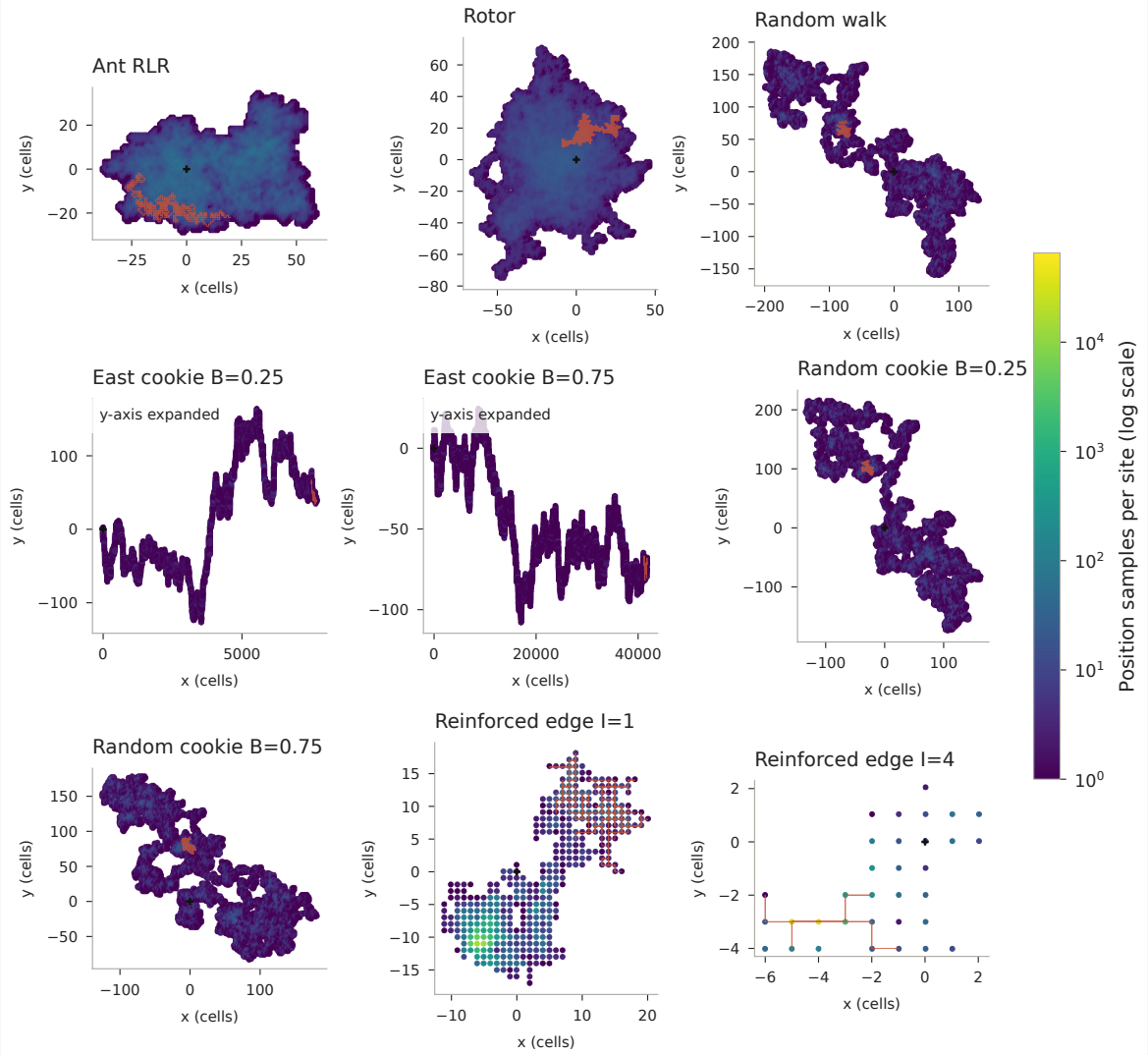


Figure 11: Nine walkers after the same 65,536 steps. B labels the once-excited (cookie) walks and I the linearly edge-reinforced walks described in the text above. Stochastic walkers use the preselected seed 301; the ant starts blank. Colour records position samples per site (including the origin); orange traces the final 1,024 steps and + marks the origin. Panels have equal x/y scaling except the directional-cookie panels, whose y-axis is expanded to show transverse motion. Spatial scales differ: read the cell coordinates. Directional cookies spread east, random cookies resemble the random walk, and strong reinforcement repeatedly uses a small territory. These single trajectories illustrate behaviour, not the ensemble exponents. B is the first-departure probability of following the cookie instead of choosing a uniform neighbour; I is the edge-weight increment.

The registered pilot used eight seeds (301–308) per setting, up to 2^{20} steps on an 8192^2 board. Table 4 reports per-seed fits over $k = 14$–18 and κ at 2^{18}; uncertainties are between-seed sd. All 16 directional-cookie runs escaped before completing that window, so we report no fitted exponent for them. Random-arrow cookies stayed close to the random-walk control. Weak edge reinforcement had a mean area exponent near $\frac{2}{3}$, but its large scatter and falling κ do not establish the ant regime. Strong reinforcement greatly reduced exploration.

Walker	Area slope a	κ at 2^{18}
Random walk	0.924 ± 0.043	1.350 ± 0.082
Rotor	0.740 ± 0.027	1.845 ± 0.118
Random cookie, $B = 0.25$	0.942 ± 0.047	1.327 ± 0.075
Random cookie, $B = 0.75$	0.920 ± 0.044	1.293 ± 0.094
Edge reinforcement, $I = 1$	0.698 ± 0.303	0.312 ± 0.132
Edge reinforcement, $I = 4$	0.187 ± 0.092	0.328 ± 0.447

Table 4: Stochastic memory pilot, eight complete seeds per row. The early rotor area slope overshoots $\frac{2}{3}$ by 0.074, so this window cannot identify an asymptotic exponent. In the later $k = 16$–20 window, the rotor gives 0.715 ± 0.019, weak edge reinforcement 0.701 ± 0.277, and strong reinforcement 0.216 ± 0.096. Weak reinforcement's mean κ falls to 0.155 ± 0.084 at 2^{20}.

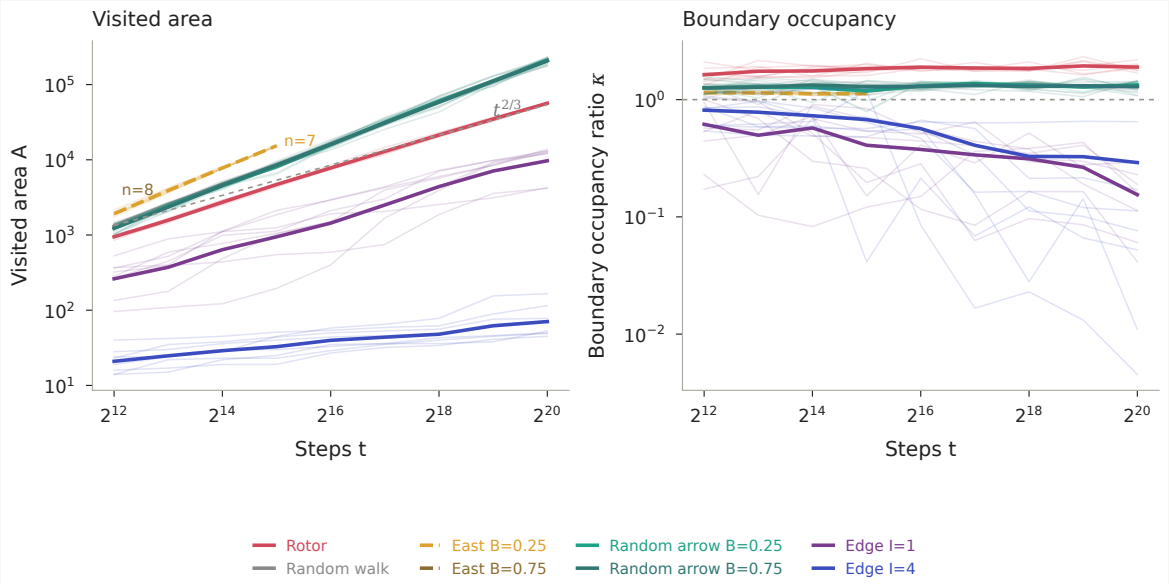


Figure 12: Growth and boundary occupancy across the same pilot. Thin lines are individual seeds; thick lines show the geometric mean area and arithmetic mean κ. All solid curves have eight seeds. Dashed directional-cookie curves end when runs escape; terminal labels give the number still contributing, and later points describe survivors only. The grey guides show area proportional to $t^{\frac{2}{3}}$ and $\kappa = 1$. Persistent memory can suppress exploration without producing stable ant-like boundary occupancy.

These controls separate suppression of exploration from stationary boundary occupation. Random cookies at $B = 0.75$ remain near diffusive area growth, above the registered $a = 0.60$–0.75 band. Weak reinforcement enters that area band but gives κ below the registered 0.5–3 band and declining with time. Thus neither the presence of spatial memory nor an area slope near $\frac{2}{3}$ suffices to reproduce the joint ant regime in these settings. Directional-cookie escapes further bound the one-use directional-memory comparison. The pilot windows, seed variability and early rotor bias in Table 4 prevent an asymptotic exclusion. Recipes and per-seed fits are in Audit 20 and Audit 21.

Randomising rotor arrows tests the preservation of deterministic phase more directly (Audit 25 $\varepsilon = 0$–0.1, eight seeds per setting, 38 of 48 runs fitted over $k = 20$…26). The noiseless rotor gives $a = 0.69$, $\kappa = 2.0$; at $\varepsilon = 0.001$, $a = 0.76$, and at 0.03 it reaches 0.94. Boundary occupation responds non-

monotonically: κ rises to 3.6, 5.8 and 6.7 at $\varepsilon = 0.001, 0.003, 0.01$, with positive temporal drift, before falling at larger noise.

These crossover values exclude the registered smooth interpolation towards the memoryless value 1.4 over the measured windows. Meanwhile c remains 0.498. The intervention therefore separates occupation from local edge-entry bias: arrow randomisation strongly changes the former while preserving the latter. This supports phase-preserving updates as a structural ingredient in the rotor comparison, without establishing their necessity for every ant-like process.

5.5.3 Shallow trapping does not account for occupation differences

Layer profiles bound explanations based on shallow trapping (Audit 22 prior predictions in Audit 23). Layer d is Manhattan distance to never-visited ground minus one; $\rho(d)$ is visits per cell relative to the territory mean, with $\rho(0) = \kappa$. In 24 selected blank-board ant rules (12 extreme and 12 near $\kappa = 0.9$; Audit 24), 15 have depleted layers 2–4 ($\rho = 0.01$–0.6). In 13 of those, $c = 0.61$–0.66 and edge-phase dwell is 1.5–1.6 visits per cell, despite a crust-class κ range of 0.13–18. Boundary occupation thus varies widely among rules with similar local edge kinetics.

The other nine have deeper occupation within 30% of uniform and $\kappa = 0.82$–0.96, a range narrowed by selection.

The class split is exploratory. Depletion extends beyond layers 0–1, and dwell–occupation correlations are negative (Spearman $\rho = -0.26$ in the extreme cohort, -0.59 in the typical cohort), constraining both the registered first-two-layer explanation and the proposed positive dwell association.

The matched comparison further limits crust depth as an ordering variable (Audit 25): among 26 pairs matched on length, R count and edge-entry probability within 0.02, with rule-mean κ differing by at least 0.5, the lower-κ rule has the deeper crust in only 13 pairs, below the registered 18. Blank-board occupation preserves 25 of the 26 patched-start pair orderings. The reproducible occupation differences therefore require information beyond these local dwell and shallow-depletion summaries. Interior return dynamics are a candidate source of that information; the local tests alone do not establish their role.

5.5.4 Interior-return associations and their size confounds

Excursion tomography bounds which measured return summaries can account for κ beyond territory size (Audit 13 analysis and provenance in Audit 14). Apparent associations involving return displacement divided by territory radius and return angle were largely attributable to territory size. Early excursion features supplied no held-out predictive gain for later $\log \kappa$ beyond an area baseline, limiting these summaries as independent transport predictors. Extreme completed-excursion duration relative to territory radius remains associated with κ, but the ratio retains a size confound, its binned quantile has coarse resolution, and window-crossing excursions are censored. The rotor control was outside its registered tolerance and final-window random-walk controls were unavailable, limiting the strength of the exclusion. These measurements narrow the candidate explanations without establishing a mechanism; matched-area comparisons or interventions are needed.

A subsequent registered analysis assigned area matching, κ and the excursion cutoff to three disjoint seed groups, rotating their roles (Audit 15). Across 44 previously selected rules, the rule-bootstrap intervals for pairwise concordance excluded chance at both measured windows. This weakens shared-trajectory area selection as a sufficient explanation.

A registered tail-budget test then asked whether the mean duration of interior returns predicts the boundary share beyond boundary residence and edge geometry (Audit 43 44 selected rules, 32 fresh

starts each, three area bands). With the empirical binned mean duration as the estimator, the leave-one-rule-out gain was 0.35 at the smallest band but its rule-bootstrap lower bound was −0.02, so the registered criterion failed; at the second band the gain was 0.51 with lower bound 0.25. Extrapolating to the largest band met the registered error threshold (median relative error 0.12 against 0.20), but a model-free log-linear extrapolation of the outcome seeds did better (0.07) and a 99.9th-percentile summary did worse (0.5). The pass therefore does not support the tail. A Shapley decomposition of the model change between bands attributes 100% of the change to the tail terms, but this follows largely from the identity (the mean duration far exceeds boundary residence) and is a model counterfactual, not causal evidence. The analysis scripts were written after the run with their choices fixed beforehand, the prediction without a perimeter-to-area rescaling is our choice, and the survival-family fit was not frozen. This is one selected cohort and no census-wide claim follows.

It is an association on existing simulations, not fresh-rule prediction: matching noisy area still leaves size uncertainty, escapes are excluded, and the two windows reuse trajectories. The earlier matched-area comparison did not meet its registered noise-floor gate (Audit 16), so it remains insufficiently resolved for the deferred transplant. The later concordance neither changes that gate outcome nor establishes a causal cutoff mechanism.

Together these constraints require an explanation beyond the tested local reflection, generic-memory and shallow-trapping closures. They focus attention on the evolving board state and interior returns, while leaving both the longer-prefix prediction and the residual cutoff association to be explained.

5.6 Growth outliers accompany drifting edge occupation

Slower and faster growth track how often the ant reaches the edge. The entry probability is comparatively stable; the relative time spent at the boundary changes.

425 rules grow logarithmically (α median 0.101), with κ median 0.001 and $\gamma_\kappa = -0.79$: the ant rarely visits the edge and the occupancy of the edge decays, consistent with an ant confined near its origin.

109 blob rules have $\alpha > 0.37$ with γ_κ median 0.17 (edge occupation rises). Across blob rules the correlation of α with γ_κ is 0.78, with γ_c 0 and with β 0.18 (Figure 13).

These deviations locate the limit of the stationary-factor description: the outlying growth exponents accompany drifting edge occupation, the departure specified in advance ("κ drifting strongly with t"). Through Equation 2 this identifies which measured factor carries the deviation; the physical origin of its drift remains open.

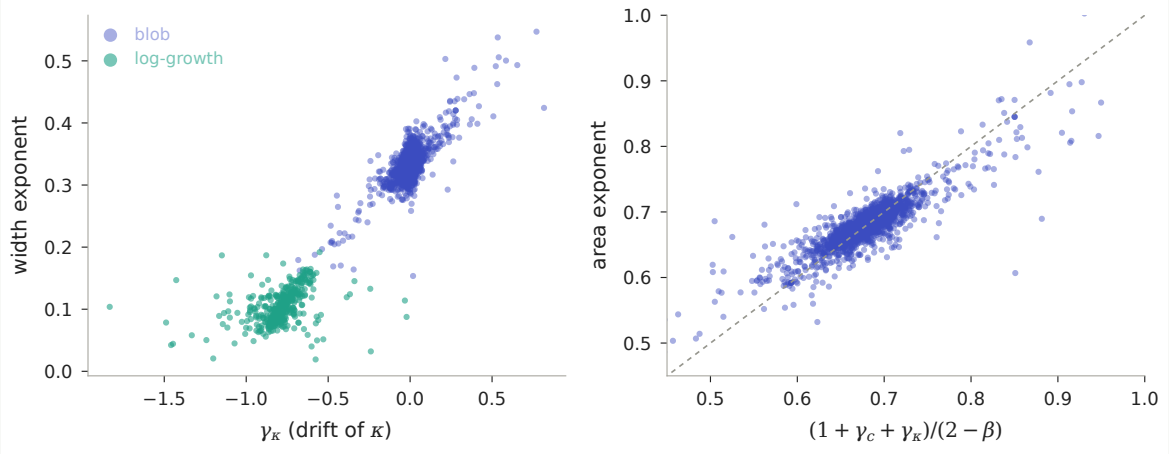


Figure 13: Left: width exponent against the drift exponent of κ; shared observables enter both axes. Right: area exponent against its reconstruction from Equation 2; the agreement is by construction up to fit error (see text). Neither panel tests a mechanism.

6 Predicting edge entry and testing interior mixing

6.1 Rule letters predict edge-entry probability

Young edge colours encounter fresh ground more often on the left. For edge steps on colours 1 to 3, the share with fresh ground on the left exceeds that on the right in 294 of 300 calibration rules (medians 0.54 against 0.17). This handedness, or chirality, means that a rule turning L on a young colour enters new ground more often.

Across rules c ranges from about 0.1 to 0.67. A chirality model uses the rule letters and this measured asymmetry to predict c. This is the model behind the abstract's score. The 0.78 quoted in Section 5 comes from a plain linear regression on string features, with different features, sample and split; the two R^2 values are not comparable.

Constants (the share of edge steps by colour and the left/right fresh-ground rates) are fitted on a 300-rule sample and applied to the others without refitting.

On 2747 held-out rules the model attains $R^2 = 0.66$ (correlation 0.87 median absolute error 0.04); grouping rules by their first letter after the leading R explains 0.59 of the variance of c, by the next three letters jointly 0.81, and by the sixth letter alone (rules of length at least 8) 0.001.

The model was a post-hoc fit to these data, though evaluated on rules outside its calibration sample; we have not tested it on rules of length above 12.

Comparison with letter-group baselines. A critique-driven reanalysis retains all 2747 held-out rules and fits group means only on the same 300 calibration rules. The primary baseline groups by the letter after the leading R; the secondary baseline groups by the next three letters jointly, with a training-only fallback for absent groups. The groups and scoring procedure were fixed before this comparison, after the original scatter plot had been inspected.

Prediction	Held-out R^2	RMSE
One-letter group mean	0.586	0.081
Three-letter group mean	0.808	0.055
Chirality model	0.66	0.074

Table 5: Predictions of c on exactly the original held-out rules. Group means use calibration targets only. R^2 uses the held-out-mean denominator, matching the original model score; using the training-mean predictor as reference gives 0.587, 0.808 and 0.661, respectively.

Chirality reduces squared error by 18% against the one-letter baseline (95% interval 15.9–20.1%). It has 76.7% *more* squared error than the three-letter baseline (interval 65.4–88.9%). These are paired rule-bootstrap intervals with 4000 draws, seed 20261046, stratified by the primary groups and conditional on the fitted models; related rules and initial-seed variability limit their interpretation.

Within the broad L and R groups, chirality still has $R^2 = 0.19$ and 0.164, with 1452 and 1295 held-out rules. It therefore has some discrimination beyond one-letter grouping. After subtracting each three-letter group's held-out mean from both observations and predictions, however, its pooled within-group R^2 is approximately zero (−0.00006); the corresponding raw within-group score, which also penalises group-mean errors, is −0.776. Thus the three-letter lookup is the stronger predictor, and the chirality model supplies essentially no additional residual discrimination within those groups. This observational comparison does not explain the growth exponent. Reproduction and per-group scores are recorded in Audit 42.

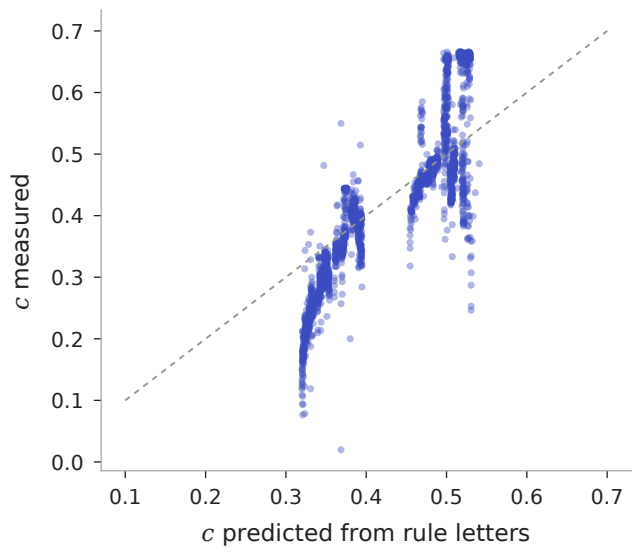


Figure 14: Held-out edge-entry probability: model against measurement. Predictions concentrate near 0.33 and 0.53 (rules with predicted $c \approx 0.53$ have measured c from about 0.25 to 0.67). The letter-group comparison in Table 5 quantifies the limitation: modest skill remains within one-letter groups, but essentially none after centering within three-letter groups.

6.2 Interior colour uniformity and transport tests

A candidate interior approximation treats the ant's turns as those of a persistent random walk with turning probability $q = \#R/n$. This does not specify occupation of the evolving territory or its

boundary. The sample is the 49 rules in Audit 26, of which 33 are blob rules with $0.30 < \alpha < 0.37$; the procedure used to choose the list was not recorded, so it should be read as a convenience sample.

We therefore repeated the test on 40 rules drawn uniformly at random (fixed seed) from the core class, with predictions written beforehand (Audit 27): 35 of 40 have near-uniform interior colours (our prediction of 60–85% was exceeded), their MSD-slope ratio has median 1.01 (range 0.84 to 1.58), and the remaining 5 have median ratio 1.3.

The convenience-sample numbers follow. For the 33 typical rules we tested (a) colour uniformity (total-variation distance of interior colours from uniform), (b) turn correlations, and (c) the slope of mean-square displacement against lag, which for independent turns is $\frac{1-a'}{1+a'}$ with $a' = (2q-1)^2$.

For 24 rules with near-uniform interior colours (TV < 0.05) the measured slope is within [0.95, 1.54] of the prediction (median 1.04), the displacement exponent is 1, and the lag-1 turn correlation is −0.002 at the median.

For the other 9 rules, whose interior colours are far from uniform, the MSD slope is off by a factor of 1.32 at the median (correlation between the ratio and the colour deviation across the sample: 0.69).

Time to mix across the territory is a fraction 0.002 (median) of the growth time.

For 73% of the sampled rules, these particular interior statistics are compatible with the persistent-walk approximation. They do not establish a random-environment description of the full dynamics, uniform spatial occupation, or stationarity of κ. Boundary occupation and its drift must be measured separately; Figure 10 reports those measurements for six rules.

7 Longer runs: exponents, visit profiles and surface width

Three further measurements (Audit 28, Audit 29, Audit 30 hypotheses and predictions are in Audit 31, written before each run), bear on the weakest points above.

7.1 Longer ensembles do not resolve a universal exponent

Eleven rules (stratified by length 3–12) were run from 251 random 5×5 starting patches, 24 seeds each, to 2^{30} steps. The local area exponent, averaged over seeds and over the last six three-octave windows, has mean 0.683 across rules (sd 0.014, range 0.66 to 0.709).

The same estimator applied to 24 rotor seeds, whose exponent is proved to be $\frac{2}{3}$, gives 0.678: even the rotor sits 0.01 above $\frac{2}{3}$ at 2^{30} by this estimator. 6 of the 11 rules are within 0.006 of the rotor; the others are up to 0.02 above it or 0.017 below it by amounts many standard errors wide.

One rule (RRLLLLLLR) does not self-average: its seed-to-seed spread rises with time. So the data are compatible with a common exponent $\frac{2}{3}$ with slow, rule-dependent convergence, but they cannot distinguish that from rule-dependent exponents within about 0.02.

7.2 Ant and rotor visits accumulate in the same cone

The time-averaged visit density, as a function of distance from the centroid in units of the territory radius $R_{\text{eq}} = \sqrt{A/\pi}$ and of t/A, is a cone, $n(r) = 3\big(1 - r/R_{\text{eq}}\big)t/A$, for ants and rotor alike. Over the first ten of fifteen radial shells the ants' profile (88 runs of 11 rules, 2^{28} steps) and the rotor's (40 seeds, 2^{26} steps) differ by at most 5%, and the ant profile differs from the cone by at most 5%; the across-rule relative sd at mid-radius is 4%. This agreement bounds the bulk-profile differences despite the distinct microscopic updates, contrary to the registered prediction of differing profiles. It is not independent evidence for the cube root, because the cone is conditionally implied by the exponent

together with approximately uniform instantaneous occupancy and compact circular growth with coherent discovery radii.

Suppose the walker is spread uniformly over the current territory, $n(r,t') = 1/A(t')$ for $r < R(t')$, and $A(t) \propto t^a$. A cell at radius r is inside from the time t_r at which $R(t_r) = r$, so its accumulated visits are $\rho(r) = \int_{t_r}^{t} dt'/A(t') = (t/((1-a)A))\big(1-(r/R)^{2(1-a)/a}\big)$. For $a = \frac{2}{3}$ this is exactly $(3t/A)(1 - r/R)$, and it integrates to t over the disc. The cone is therefore the footprint of a uniformly mixing walker inside a disc growing as $t^{\frac{2}{3}}$; nothing specific to rotors or ants enters beyond these occupancy, geometry and discovery-time assumptions (Audit 19, A3). It also explains why ants and rotor agree although their κ differ (about 0.9 against 2.0): κ changes only a thin edge layer, not the bulk integral. The profile checks consistency with these joint assumptions; accumulated visits alone do not establish instantaneous uniform occupancy. Kapri and Dhar instead motivate the rotor cone by approximately equal subsequent local-time increments [7], so matching the cone does not distinguish the mechanisms.

With the measured rotor exponent $a = 0.687$ the formula gives a centre prefactor $1/(1-a) = 3.19$ rather than 3, against a measured 2.87 (Audit 29) that the 0.075 R_{eq} shell width lowers by roughly 3–4%, so the data do not separate $a = \frac{2}{3}$ from 0.687 at better than about 5%.

Controls limit the profile derivation. The formula's predictions for other exponents were tested on walkers with $a \neq \frac{2}{3}$ (Audit 19, W2, W3 and the checks after them; Audit 32, Audit 33), with results that do not support the derivation beyond $a = \frac{2}{3}$.

Rotors whose arrow is randomised with probability $\varepsilon = 0.001$ ($a = 0.75$–0.78, four seeds, territories round enough for a radial profile: anisotropy ≤ 0.31, fill 0.96) have a centre density of 1.9–2.1 against 3.5–3.8 from the formula, so the formula fails and the cone, at 3, is also off. For $\varepsilon \geq 0.003$ and for a memoryless persistent walker the territories are elongated or holey (anisotropy up to 0.9, fill down to 0.3), so the premise fails first and the profile cannot test the formula.

Two chaotic 2-state turmites with a stable $a \approx 0.74$ over six octaves (`tm:2:2:46382`, `7982`; single trajectories) keep a cone-like centre (2.8 and 3.1) below the formula's 3.5 and 3.2.

So the cone is not simply a consequence of "uniform occupancy and an exponent", and we cannot say that uniform occupancy needs $a = \frac{2}{3}$; every walker we tried has $a \geq \frac{2}{3}$, and the exponents are finite-window values.

We had also meant to test the formula on the $a \approx \frac{1}{2}$ turmite cluster, but those tables grow as one-dimensional lines (anisotropy 1.00), and eight of the ten sampled are the same trajectory, so the cluster does not bear on the question.

The territories are round (median anisotropy 0.07 at 2^{28}, the rotor's 0.054 at 2^{26}; fill 0.98).

7.3 Surface width grows faster in the examined ant cohorts

The surface width does *not* match: its local exponent fitted over $k \geq 18$ is 0.26 for ants and 0.16 for the rotor (Kapri and Dhar: about 0.13), so these fits show a finite-window temporal-exponent gap, without matching physical radii. These single fits use a different window ($k \geq 18$) from the matched-k follow-up below, which is why the absolute values differ; the comparison that matters is the ratio.

A longer follow-up (Audit 34 36 ant runs and 14 rotor seeds to 2^{34}) finds both exponents flat rather than falling, about 0.22–0.25 for ants and 0.10–0.13 for the rotor at every matched k from 22 to 32, so the gap of roughly a factor two does not close in the accessible range; whether it closes asymptotically is undecided.

The gap is not an artefact of the extreme-κ rules with a "crust" (an almost unvisited shell between the active edge and the bulk, found in 15 of 24 rules examined; see Section 5): to 2^{30} the 9 ordinary-κ rules without a crust have local exponents 0.21–0.35 (median 0.27), all above 0.18, against 0.17–0.45 (mean 0.32) for the 15 crust rules (difference +0.04, Mann–Whitney $p = 0.054$; Spearman with $\log \kappa$ +0.40, $p = 0.056$; one trajectory per rule, W1). The wider edge occurs in both examined ant cohorts, with at most a mild additional effect in this high-κ sample; census-wide generality is not established.

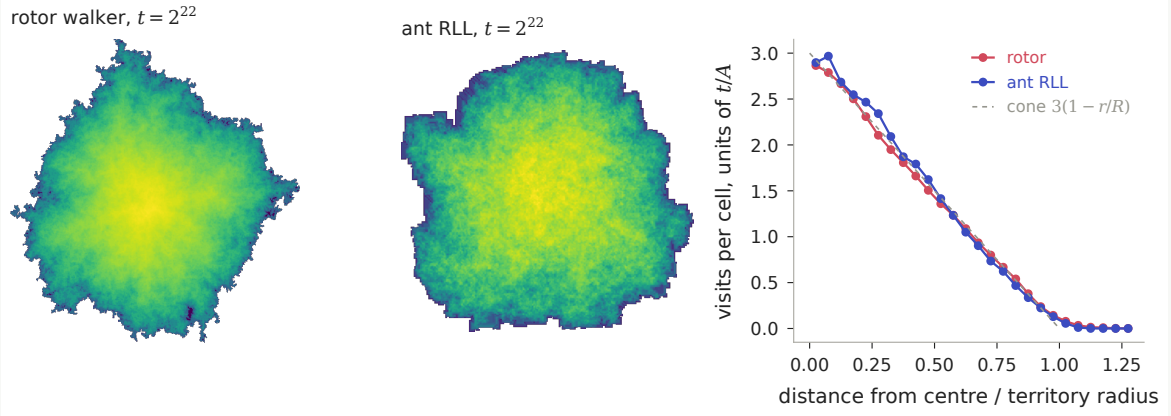


Figure 15: Left and middle: how often each cell was visited (log scale) in a single run of 2^{22} steps, for the rotor walker (one seed) and the ant RLL. Right: visits per cell, averaged in rings around the centre, against the cone $3(1 - r/R)$. Both curves follow the cone; this figure is one run each and is an illustration, not the ensemble measurement quoted in the text.

7.4 The width gap survives radius and outer-hull checks

Refitting the same A4/A4c trajectories directly against $R = \sqrt{A/\pi}$ gives $W_{\text{surf}} \propto R^{\zeta}$ with rotor $\zeta = 0.401$ (14 seeds, sd 0.073; bootstrap mean interval 0.366–0.437) and ant $\zeta = 0.714$ (36 runs across six rules, RMS within-rule seed sd 0.134; rule-bootstrap interval 0.664–0.761), over $k = 22$–32. The rotor estimate lies above the registered $\zeta \in [0.30, 0.37]$ band and is closer to $\frac{3}{8}$, implied by $W \propto t^{\frac{1}{8}}$ and $R \propto t^{\frac{1}{3}}$, than to $\frac{1}{3}$. Restricting to $k = 26$–32 gives 0.389 and 0.685; the later A4c cohort alone over $k = 26$–34 gives 0.395 and 0.637. The ant/rotor slope ratio therefore changes from 1.78 to 1.61.

Replacing R with R_g or the mean outer radius changes these means by less than 0.005. This confirms a finite-window gap but does not support an exact factor-two law.

The surface cells here, as in the edge P of Equation 1, are all visited cells with a never-visited 4-neighbour, so they include the rims of interior holes and detached unvisited pockets, not only the outer hull; W_{surf} is the standard deviation of their radius about the centroid (the same surface-site criterion as Kapri and Dhar, but they use origin-centred radii and the square root of an ensemble mean of per-cluster variances; we summarize per-run fitted slopes). Their v2 body gives 0.136 ± 0.02, while Fig. 8 gives ± 0.002; we therefore quote only about 0.13 [7].

Interior holes do not explain the gap. A follow-up separates hole rims from the outer coastline (24 ant runs of the six rules above and 8 rotor seeds, to 2^{30}; Audit 35, predictions registered before the run). Porosity does not explain the gap: the local exponent over $k \geq 22$ is 0.239 for the ants' outer-hull width and 0.239 for all surface cells, against 0.135 and 0.130 for the rotor, and hole rims are 4% of the ants' surface cells but 25% of the rotor's.

The ants have a lower finite-resolution box-count slope (1.07 against 1.16) and steeper measured spectral decay than the rotor; these are not established asymptotic fractal dimensions.

The wavelength follow-up finds relatively greater low-mode excess, but its registered band decision is mixed and the high-band result is outside the predicted range. The temporal fall in normalized power includes $\left(\frac{W}{R}\right)^2$ and supplies no independent corroboration. Persistence of large-scale shape versus local fluctuating width remains unresolved (Audit 35).

These are growth-trajectory slopes, not spatial roughness exponents; physical-distance height correlations, local widths, spectral scaling and phase persistence beyond the already removed modes 0–2 are needed, together with a justified growth clock, to test a KPZ or other interface universality class [13], [14] (Audit 33).

7.5 Replicated starts preserve extreme growth exponents

We re-ran the 36 fastest and 32 slowest blob rules from 12 random patches each. If their exponents had been run-to-run noise they would regress to the middle. They mostly do not: only 17% of the fast rules fall below 0.37 and 34% of the slow rules rise above 0.30; the Spearman correlation between census and re-measured exponent is 0.84, and the re-measured exponent correlates with γ_κ at 0.82. So the outliers are real properties of the rules over 2^{20}–2^{27} steps, with edge occupation drifting; whether they are different asymptotes or very slow transients is untestable at this length (the exponent comparison shows even typical rules differ by about 0.02 at 2^{30}).

The replication bounds the noise interpretation to typical rules, narrowing our earlier statement about between-rule scatter: there it is (within-rule sd 0.016 against a between-rule sd of 0.004), but the tails are not.

7.6 A blob becomes a column sweeper after billions of steps

A long blob-like transient can end in a different growth regime. In one blank-board trajectory, RLRLLL ran as a blob until step 1.2677×10^{10} and then switched within a few 10^6-step windows to a steady builder at 4.02 steps per newly visited cell (all 299 later windows), and left a 32768^2 board at 1.2974×10^{10} (Audit 38, Audit 39 one blank-board trajectory, trigger unknown).

The new regime sweeps columns back and forth. Two 4×10^5-step trajectory dumps at 1.28×10^{10} and 1.29×10^{10} (Audit 40, Audit 39 Audit 41) show that this builder is a boustrophedon column sweeper rather than a perimeter crawl: it makes a slow traverse of the full height L (6,400 rows at the first dump and 8,610 at the second) at about 10 steps per row, a fast return at about 2 steps per row, and adds three columns per round trip, and the wake is exactly periodic in x with period 3 (column agreement 0.99, colours 3, 4, 5 only).

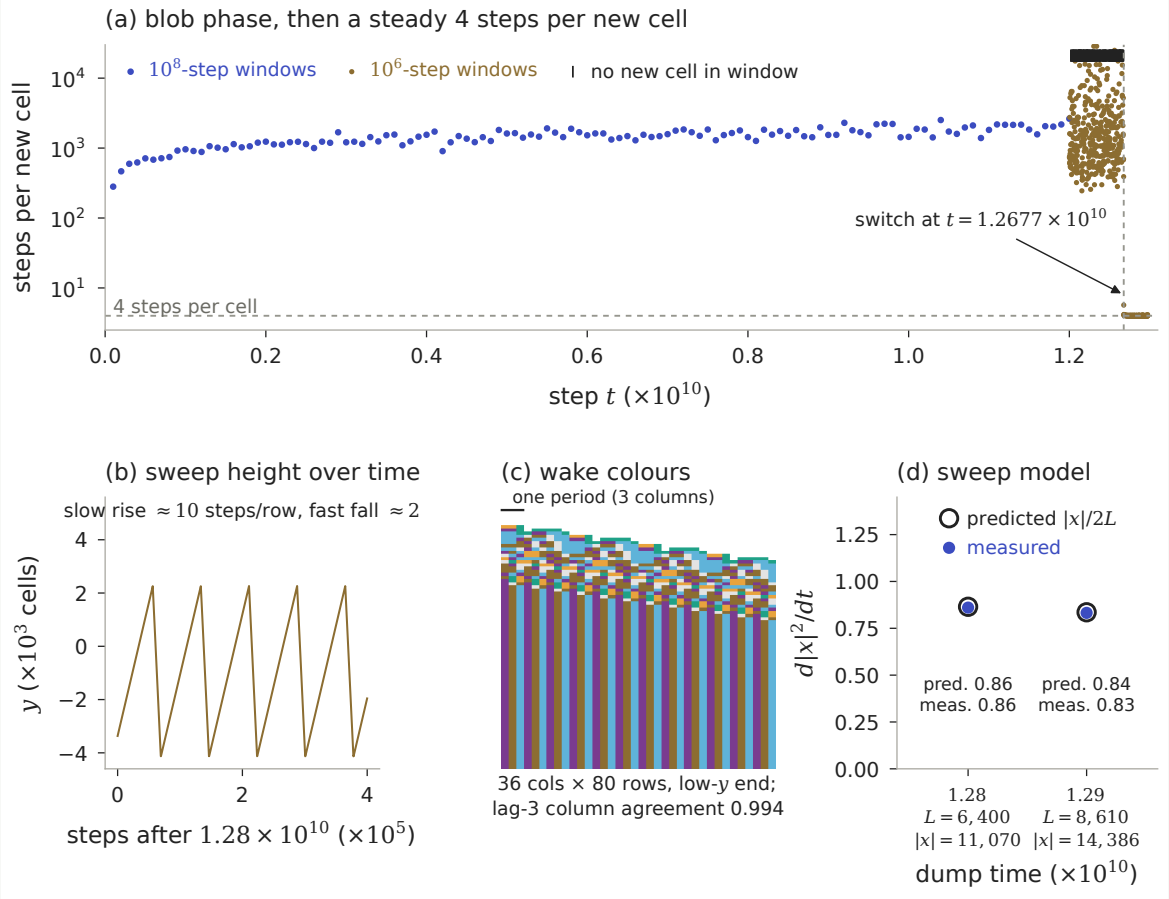


Figure 16: The RLRLLL switch in one blank-board trajectory (a single run, no replicates). (a) Steps per newly visited cell against t, in 10^8-step (coarse) and 10^6-step (fine) windows; black ticks mark fine windows with no new cell; the guide marks 4 steps per cell and the switch time. (b) The ant's row $y(t)$ over a 4×10^5-step dump at 1.28×10^{10} steps: slow traverse, fast return. (c) A stretched 36×80 crop of the period-3 wake from the same dump, with colours 3, 4, 5 drawn in the palette. (d) Predicted ($|x|$ /$2L$) against measured $d|x|^2$ /dt at the two dumps; both L come from the same trajectory, so this is a consistency check on the sweep picture, not a test of $L \sim |x|$.

Three columns of L cells per $\approx 12L$ steps (Figure 16) is 4 steps per new cell, and $dx/dt = -1/(4L)$ with the measured $L \approx 0.67\ |x| - 960$ predicts $d|x|^2\ /dt = 0.86$ and 0.84 at the two dumps against 0.86 and 0.83 measured, so $|x|$ grows as $t^{1/2}$ in this phase.

The two values of L come from the same single trajectory, so this is a consistency check on the sweep picture, not a test of the $L \sim |x|$ law; why the pattern has period 3 and what triggers the switch are unexplained. Three further board escapes without a recognised highway in the 10^{11} sample may be of this kind but were not traced.

8 Evidence summary: tested claims and open mechanisms

Claim	Status	Evidence
Width $\sim t^{\frac{1}{3}}$, area $\sim t^{\frac{2}{3}}$ for non-highway rules	Measured; consistent after the rotor-calibrated bias	Table 2, Figure 9
c constant in time	Measured drift is small over the tested windows	Figure 10, γ_c
κ constant for typical rules	Drift below 0.05 for 83% of blob rules; systematic drift in log and fast classes	Figure 10, Figure 13

Claim	Status	Evidence
$P \sim A^{\frac{1}{2}}$	Finite-window constraint: measured $\beta = 0.525$, equal to the rotor's under the same window	Table 2
Equation 2 reconstructs a for outliers	Conditional scaling algebra; fit consistency only, not a test of mechanism	Figure 13 (right)
Edge-entry probability c from rule letters	Held-out $R^2 = 0.81$ for three-letter groups, 0.66 for chirality; length ≤ 12 only	Table 5; Figure 14
Interior persistent-walk approximation	Compatible colour and short-lag statistics for 24 of 33 convenience-sampled rules and 35 of 40 random core rules; spatial occupation and boundary access are not established by these tests	Mixing paragraph
Occupation profile is the rotor's cone	Measured within 5% of the rotor, excluding the registered profile separation at that scale. Derived conditionally from uniform instantaneous occupancy, compact circular growth with coherent discovery radii and $A \sim t^{\frac{2}{3}}$, so not independent of the cube root. **Unexplained**: why the profile is the cone. The derivation from uniform occupancy plus the exponent fails for walkers with $a \neq \frac{2}{3}$ (noisy rotor, random-background ants), while two turmites with $a \approx 0.74$ keep a cone-like centre	Section 7
One exponent for all rules	Neither confirmed nor refuted: rules differ by up to 0.02 at 2^{30}, as does the rotor from its proved value	Section 7
Outliers are noise	Replication preserves most extreme exponents: only 17% / 34% regress	Section 7
Surface width matches the rotor	Not supported: ants' exponent is about twice the rotor's (0.26 against 0.16 for a fit over $k \geq 18$; 0.22–0.25 against 0.10–0.13 at matched k = 22–32)	Section 7
κ is rule-specific and partly predictable	Moderately reproducible (ICC 0.58). Simple letter regression: $R^2 \approx 0$ for κ. Prefix-8 lookup: $R^2 = 0.633$ and gain 0.24 [0.148, 0.347] beyond early area, **on stationary test rules with stationary training**. On all-rule evaluation: gain 0.031 (interval includes zero). Interpretation: prefix predicts κ conditional on stationarity; does not predict stationarity. No mechanism established	Section 5
Why κ takes its value, why it drifts in some rules	**Unexplained**; a memoryless persistent walker (A1d) gives $\kappa \approx 1.4$ and area exponent 0.95, rejecting that memoryless control; edge-layer anatomy (A1-typical) separates a crust class from a uniform class but constrains shallow trapping: similar dwell times coexist with widely differing κ; κ is not shared between cyclic rotations of a rule (ICC 0.20, K1); a stochastic true self-avoiding walk gets κ up to 1.97 with area exponent 0.98 (E1), so $\kappa \approx 2$ is not specific to the rotor	Section 5
Same-run prefactor collapse using $c\kappa$	Descriptive: dividing by $c\kappa$ cuts the between-rule variance of $\ln\left(A/t^{\frac{2}{3}}\right)$ at 2^{26} by 59% (pre-registered level 75%); a free fit gives power 0.7, $R^2 = 0.73$. Factors share the growth data, so this is not independent validation. Geometry amplitude $\frac{P}{\sqrt{A}}$ also enters the stationary prefactor	Audit 8
Rule-specific area exponent is filling-fraction drift	Filling drift is insufficient: γ_η explains $R^2 = 0.45$ of $a - 2\alpha_w$ and has ICC 0.11 (predictions ≥ 0.80, ≥ 0.35); the two box sides co-vary instead	Audit 7

Claim	Status	Evidence
Whether the edge exponent is 1/2 asymptotically	**Unexplained**; the finite-window β is 0.525 for ants and 0.523 for the rotor; to 2^{36} the ants (six rules, two starts each) read 0.509 ± 0.003 against the rotor's 0.501 ± 0.003, both close to 1/2, but the decision rules we set in advance are inconclusive (W4)	

9 Limits of finite-time scaling and regime classification

- **One trajectory per rule.** The ant is deterministic from a blank board; the only noise estimate is the patched-start control (40 rules, 474 runs). Rules are not independent samples of anything; the bootstrap interval is a statement about this set of rules.
- **Finite windows.** All exponents come from seven octaves; the rotor calibration quantifies but does not remove the bias, and the rotor's bias need not equal the ants'. Long runs to 2^{32} cover only six rules.
- **Rule length ≤ 12.** Both the census and the c model stop there; the prefix-κ test extends to sampled lengths 13–14 under core-band and stationarity filters, not to an exhaustive growth census. Rules with large n are the majority of all rules; if the cube-root regime is a property of short rules, we cannot tell.
- **Edge exponent at finite time.** A separate check on the six long runs (Audit 36) gives local boundary exponents of P against A (slope over five octaves from k; a different estimator from the census β, hence 0.51–0.53 rather than 0.525), averaged over the six rules, of 0.53 at $k = 20$, 0.51 at $k = 26$ and 0.52 at $k = 28$, with single rules between 0.48 and 0.55 and no clear fall; the rotor's local slope is 0.52 at $k = 22$. A pure $A^{\frac{1}{2}}$ law fits worse than $P = c_1\sqrt{A} + c_0$ (rms log-residual 0.045–0.16 against 0.018–0.073 over the six rules), so the data are consistent with a square-root edge plus an offset shared by both walkers, but do not establish the asymptotic value. Extending the six ant rules (blank and one patched start each) to 2^{36} gives a mean local slope of 0.509 ± 0.003 (twelve runs; the rotor at 2^{36}: 0.501 ± 0.003), with a fall of 0.009 ± 0.004 over the last two octaves against the rotor's 0.012 ± 0.004; our pre-set rules for "consistent with 1/2" and "plateau above 1/2" both return inconclusive, and the fall again rests on the last octave.
- **Class boundaries.** 153 rules were not fitted. The builder and log classes are operational definitions with thresholds (20 steps per box cell, $\alpha < 0.15$); the log cut is checked against the exponent distribution in Section 2.1.
- **Highway builders.** Classification is by finite simulation. A rule counted as chaotic may build a highway after 2^{27} steps (Langton's ant itself takes about 10^4). Several hundred rules of length ≤ 10 were left unresolved in an earlier highway census (Audit 1). Of 807 such rules run to 10^{10} steps, 38 (4.7%) became highways in $(10^9, 10^{10}]$. In a 240-rule sample run to 10^{11} only 3 (1.25%) did in $(10^{10}, 10^{11}]$, so the tail is thinning, not flat, but one decade cannot show that the set of late highways is finite (Audit 37 the other 492 survivors were not run in that sample). The late column-sweeper transition in Section 7.6 shows that even a long blob-like transient need not persist.
- **No dynamical closure.** The factorization is exact accounting. Stationary c, stationary κ and $P \sim \sqrt{A}$ conditionally imply the cube root, but we do not derive these conditions from the rule. In particular, the rule-specific level and drift of occupation remain unexplained. The area-budget forecast tests temporal persistence without identifying the separate factors.
- **Related but untested ideas.** Experiments in which the ant's territory was replaced by statistically matched fields (pair, 2×2, 3×3) suggest that the log-growing "tethered" rules depend on the exact configuration of the territory rather than on local statistics. That work was run on two rules at one age with one trajectory each, and a follow-up knockout map has not been run. We draw no conclusion from it in this paper.

10 Conclusion: calibrated growth, unresolved transport

Most measured blob rules of length 3–12 with no detected highway within the simulated horizon grow, to within the bias our calibration measures, as $t^{\frac{1}{3}}$ in linear size and $t^{\frac{2}{3}}$ in area. The growth rate splits into an edge-entry probability and an edge-occupation ratio with small measured drifts for the large majority of rules, together with an edge that scales close to $A^{\frac{1}{2}}$; departures from the cube root accompany drifting edge occupation (correlation of α with γ_κ 0.78 over blob rules). This is bookkeeping through Equation 2, not a cause or independent validation. Rule-letter models predict part of the variation in c and, conditionally, κ; they do not derive the stationary transport regime, its occupation level, or its geometry. The contribution is a calibrated growth census, predictive regularities and constraints on tested mechanisms; the mechanism of cube-root growth in these ants remains unresolved.

We did measure the factors for the rotor with arrows initialised with a northward bias (Audit 19, C2b): κ rose from 2.01 to 2.20 at bias 0.15 while the width exponent stayed at 0.32–0.33, until the cube root broke between bias 0.25 and 0.30; randomising the arrow with probability ε raised κ up to about 3-fold and destroyed the cube root (Arm B). Together these interventions separate occupation level from the growth exponent and local entry bias, constraining any proposed explanation of the ants' rule-specific κ.

A broader candidate remains to treat κ as a hitting problem, although the excursion summaries tested above did not explain its rule dependence beyond territory size: uniform marginal interior colours alone do not specify the joint heading/colour transport law (Mixing), so κ may reflect its return time to the bulk or its hitting distribution on the boundary given the shape of the outer envelope, or a combinatorial invariant of the permutation the rule induces on a periodic lattice. We make no prediction from this here.

Meng and Cohen's earlier pocket/return argument assumes $A \sim R^2$, $B \sim R^d$, scale-independent mean pocket gain and a bulk-return-dominated collision interval $\tau \sim \frac{A}{B}$, giving $d\frac{A}{d}t \sim \frac{B}{A}$ and $A \sim t^{\frac{2}{4-d}}$ [8]. Their $d = \frac{4}{3}$ gives $\frac{3}{4}$; our measured near-$\sqrt{A}$ boundary and stationary $c\kappa$ instead give $\frac{2}{3}$. The shared scaling picture does not derive either the return law or our rule-specific occupation ratio.

The literature comparison prioritizes complete boundary/interior time budgets at fixed area and the registered hull-correlation checks before another broad census extension. Approximately balanced exploration cycles, motivated by exact rotor excursion properties [15], offer a separate test of spatial local-time increments, not an established ant mechanism.

Why longer rule prefixes carry predictive information about κ under stationary training remains open. Direct fixed-area sampling and paired mutations could distinguish shared transport dynamics from size, geometry and regime-selection effects; the prefix lookup alone cannot do so. The stationarity condition itself is not explained: why some rules settle to a stable territory and others drift is a separate open question.

Earlier flipping-rotator growth studies [8], [9] mean that the contribution is this calibrated multi-colour census and its transport comparisons, rather than a claim that territory growth was previously unmeasured. The accumulated cone requires assumptions beyond the exponent, and the measured boundary gap in selected cohorts establishes no interface class.

The future-window forecast adds evidence beyond same-window reconstruction: early area-derived budgets predict later increments substantially better than an other-rule baseline in the typical replicated run. This tests persistence of the total growth budget and supplies no independent validation

of the factorization. Rare near-stalls mark the limit of that forecast. The local, memory and geometric controls further restrict candidate transport mechanisms, while leaving the physical origin of persistent rule-level transport open.

11 Data, code and author statements

Data and code availability. [PLACEHOLDER: repository URL and archived DOI (e.g. Zenodo) to be added once the working tree is committed.] The repository contains the simulator, analysis code, numerical summaries and figure recipes. The appendix audit gives exact locations, seed records, hypothesis titles and available commit timestamps, and identifies limitations in the provenance record.

Use of AI tools. The code, analyses, and draft of this paper were produced with substantial assistance from Claude (Anthropic) via Claude Code and Google Gemini via Antigravity, under the author's direction. Codex (OpenAI) also assisted with research planning, hypothesis development, and critical review of proposed experimental designs. The author is responsible for the content and has reviewed the results. [Check wording against the target venue's policy.]

Competing interests and funding. [PLACEHOLDER]

12 Appendix: Reproducibility & Preregistration Audit Table

This audit preserves the repository locations displaced from the main narrative. All paths are relative to the repository root. Audit numbers link the physical tests to their records; they are not additional scientific results.

Repository baseline: 71ffb99863465b7dbc8f7912571af9c32626f339 (2026-10-15T22:41:51+13:00). The manuscript and some research records contain uncommitted changes. For each record below, the Git field gives the earliest commit currently available for that path (hash and committer timestamp), not proof that the hypothesis was committed before measurement. Document dates are author-recorded dates; migration commits can be later. Missing seeds or registration evidence must not be inferred from these timestamps.

The source and recipe locations below retain commands, full seed schedules, windows, input hashes and exclusions where recorded. Seed excerpts are quoted literally from those sources; they may describe sampling or bootstrap seeds rather than walker seeds. Blank-board deterministic runs need no random seed. "Not located" means this audit did not recover an explicit seed declaration, not that a run was unseeded.

Audit	Exact records and producers	Seeds, registration and Git record
1	experiments/2026-10-13-census-sweeps/results/sweep_v2.csv experiments/2026-10-13-census-sweeps/scripts/census.py experiments/2026-10-13-census-sweeps/results/unresolved_le10.txt experiments/2026-10-13-census-sweeps/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-census-sweeps/scripts/census.py experiments/2026-10-13-census-sweeps/scripts/equiv_census.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
2	experiments/2026-10-13-probe-growth-census/results/growth.csv experiments/2026-10-13-probe-growth-census/results/probe_summary.csv experiments/2026-10-13-probe-growth-census/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-probe-growth-census/scripts/c_letters.py experiments/2026-10-13-probe-growth-census/scripts/c_model.py experiments/2026-10-13-probe-growth-census/scripts/cube_fill.py experiments/2026-10-13-probe-growth-census/scripts/cube_stats.py experiments/2026-10-13-probe-growth-census/scripts/sq_baseline_window.py	Seed declarations: rng = np.random.default_rng(20261012); idx = rng.permutation(len(c)); folds = np.array_split(idx, 5) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
3	experiments/2026-10-13-b4-unfitted-rules/results/b4_builders.txt experiments/2026-10-13-b4-unfitted-rules/results/b4_stats.txt experiments/2026-10-13-b4-unfitted-rules/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-b4-unfitted-rules/scripts/b4_builders.py experiments/2026-10-13-b4-unfitted-rules/scripts/b4_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
4	tests/ant.rs Analysis scripts (commands in recipe): tests/ant.rs	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence):

Audit	Exact records and producers	Seeds, registration and Git record
		No separate registration title located in the inspected records. Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
5	tests/probe.rs Analysis scripts (commands in recipe): tests/probe.rs	Seed declarations: let seed = 7; Registration/context records (combined result notes are not independent preregistration evidence): No separate registration title located in the inspected records. Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
6	research-outputs/cube-root-growth-paper/compute_numbers.py research-outputs/cube-root-growth-paper/numbers.json research-outputs/cube-root-growth-paper/README.md research-outputs/cube-root-growth-paper/figures.py Analysis scripts (commands in recipe): research-outputs/cube-root-growth-paper/compute_numbers.py research-outputs/cube-root-growth-paper/kappa_prediction_numbers.py research-outputs/cube-root-growth-paper/figures.py	Seed declarations: rng = np.random.default_rng(20261012) uv run research-outputs/cube-root-growth-paper/compute_numbers.py # data/* -> research-outputs/cube-root-growth-paper/numbers.json (bootstrap seed 20261012) in the script (reference ant, rotor with numpy seed 1) or read from `experiments/2026-10-13-census-sweeps/results/sweep_v2.csv` and `experiments/2026-10-13-census-sweeps/results/census.csv`. Registration/context records (combined result notes are not independent preregistration evidence): Mixing test on a random sample of rules — Written 2026-10-12, before running. Reason: the earlier mixing sample notes/mixing-random-sample.md Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Prospective transport forecast — Registered 2026-10-13 23:35 NZDT, before computing forecasts. Base commit notes/transport-forecast-2026-10-13.md Earliest path commit: 9118bb51a3d74d01c99bcac1141a6d99674b4237 (2026-10-14T18:35:50+13:00)
7	experiments/2026-10-13-h1-lacunarity-decomposition/results/h1_lacunarity.txt experiments/2026-10-13-h1-lacunarity-decomposition/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-h1-lacunarity-decomposition/scripts/h1_decomp.py experiments/2026-10-13-h1-lacunarity-decomposition/scripts/h1_lacunarity.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
8	experiments/2026-10-13-h3-scaling-collapse/results/h3_collapse.txt experiments/2026-10-13-h3-scaling-collapse/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-h3-scaling-collapse/scripts/h3_collapse.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
9	notes/cube-root-hypothesis.md	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Why do chaotic ants grow as t^(1/3)? — Written 2026-10-10 07:45 NZDT, before any measurement beyond the census

Audit	Exact records and producers	Seeds, registration and Git record
		notes/cube-root-hypothesis.md Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)

10	experiments/2026-10-13-a1-kappa/results/a1b_icc.txt
	experiments/2026-10-13-a1-kappa/results/a1_kappa_proxy.txt
	experiments/2026-10-13-a1-kappa/results/a1c_predict.txt
	experiments/2026-10-13-a1-kappa/results/a1_edge_q.txt
	experiments/2026-10-13-a1-kappa/results/a1d_stats.txt
	experiments/2026-10-13-a1-kappa/README.md
	Analysis scripts (commands in recipe):
	experiments/2026-10-13-a1-kappa/scripts/a1_edge_q.py
	experiments/2026-10-13-a1-kappa/scripts/a1_kappa_proxy.py
	experiments/2026-10-13-a1-kappa/scripts/a1b_icc.py
	experiments/2026-10-13-a1-kappa/scripts/a1c_predict.py
	experiments/2026-10-13-a1-kappa/scripts/a1d_stats.py
		Seed declarations: Linear model on q = #R/n, n, letters 1..6 (R=1), run count, trained on a random half (seed 20261013) of the non-bursty blob rules of
		rng = np.random.default_rng(20261013); idx = rng.permutation(len(rules)); h = len(idx) // 2; tr, te = idx[:h], idx[h:]
		Registration/context records (combined result notes are not independent preregistration evidence):
		Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md`
		notes/frontier-2026-10-13-results.md
		Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)

11	experiments/2026-10-13-b2-between-rule-variance/results/b2_probe.csv experiments/2026-10-13-b2-between-rule-variance/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-b2-between-rule-variance/scripts/b2_anova.py experiments/2026-10-13-b2-between-rule-variance/scripts/b2_sample.py experiments/2026-10-13-b2-between-rule-variance/scripts/b2b_anova.py	Seed declarations: - `scripts/b2_sample.py`: B2: 100 random non-bursty blob rules (numpy seed 20261013), disjoint from the 40 rules of experiments/2026-10-13-patched-starts/results/patched.csv, 12 patched """B2: 100 random non-bursty blob rules (numpy seed 20261013), disjoint from the 40 rules of experiments/2026-10-13-patched-starts/results/patched.csv, 12 patched seeds each. rng = np.random.default_rng(20261013); pick = sorted(rng.choice(len(rules), 100, replace=False)) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
12	experiments/2026-10-kappa-rule-space/README.md experiments/2026-10-kappa-rule-space/results/w3c_stats.txt experiments/2026-10-kappa-rule-space/results/w3c_cross.txt Analysis scripts (commands in recipe): experiments/2026-10-kappa-rule-space/scripts/kcommon.py experiments/2026-10-kappa-rule-space/scripts/prefix_audit.py experiments/2026-10-kappa-rule-space/scripts/prefix_audit_notebook.py experiments/2026-10-kappa-rule-space/scripts/w0_build.py experiments/2026-10-kappa-rule-space/scripts/w1_smooth.py experiments/2026-10-kappa-rule-space/scripts/w2_stationarity.py experiments/2026-10-kappa-rule-space/scripts/w3b_analyse.py experiments/2026-10-kappa-rule-space/scripts/w3b_sample.py experiments/2026-10-kappa-rule-space/scripts/w3c_analyse.py experiments/2026-10-kappa-rule-space/scripts/w3c_cross.py experiments/2026-10-kappa-rule-space/scripts/w3c_sample.py experiments/2026-10-kappa-rule-space/scripts/w4_specs.py experiments/2026-10-kappa-rule-space/scripts/w4_stats.py experiments/2026-10-kappa-rule-space/scripts/w5_analyse.py experiments/2026-10-kappa-rule-space/scripts/w5_specs.py experiments/2026-10-kappa-rule-space/scripts/w6_analyse.py	W3c: "What sets kappa? A rule-space and mechanism programme", W3c registration block in README.md, author timestamp 2026-10-15 20:55 (+13:00), before sampling or simulation; result log timestamp 21:19 (+13:00), recorded commit da88e61. Runner: scripts/run_w3c.sh. Rule sampling seed 20261014+23; patched starts 501–508 (w3c_sample.py). The crossed training/test comparison is exploratory, not preregistered. The other seed excerpts below concern earlier stages. Seed declarations: rules do not straddle folds. Seeds: `numpy.random.default_rng(20261014)` for folds, written at `99de7c1`; other commits landed in between, none touching `data/`), seed 20261014: **Fresh sample.** 150 rules, 75 each of length n = 13 and n = 14 (numpy `default_rng(20261014 + 11)`), Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Excursion tomography: typical loops or rare journeys? (T1) — Pre-registered 2026-10-13 23:45 NZDT, before any code execution or simulation. notes/excursion-tomography-2026-10-13.md

Audit	Exact records and producers	Seeds, registration and Git record
	experiments/2026-10-kappa-rule-space/scripts/w6_checks.py experiments/2026-10-kappa-rule-space/scripts/w6_sample.py experiments/2026-10-kappa-rule-space/scripts/w7_emergence.py experiments/2026-10-kappa-rule-space/scripts/x1_prefix_icc.py experiments/2026-10-kappa-rule-space/scripts/x2_prefix_heldout.py experiments/2026-10-kappa-rule-space/scripts/x3_w3b_checks.py	Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: 35b42c749d9958419fef689d0fdfb7a52e437445 (2026-10-15T19:38:36+13:00)
13	notes/excursion-tomography-2026-10-13.md	Seed declarations: - Patched starts: seeds 0..7 (`RULE@0` .. `RULE@7`), $N = 44 \times 8 = 352$ runs. - Rotor walker: `rotor:SEED`, seeds 301..304. - Random walk: `rw:SEED`, seeds 301..304. Registration/context records (combined result notes are not independent preregistration evidence): Excursion tomography: typical loops or rare journeys? (T1) — Pre-registered 2026-10-13 23:45 NZDT, before any code execution or simulation. notes/excursion-tomography-2026-10-13.md Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: 99de7c1dacd81e2e065c7acfc46d6a19ae51370a (2026-10-14T00:16:50+13:00)
14	experiments/2026-10-14-t1-excursion-tomography/ experiments/2026-10-14-t1-excursion-tomography/README.md Analysis scripts (commands in recipe): experiments/2026-10-14-t1-excursion-tomography/scripts/t1_audit.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_depth_analysis.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_excursions.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_pair_disp.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1_specs.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1b_analysis.py experiments/2026-10-14-t1-excursion-tomography/scripts/t1b_cutoff.py	Seed declarations: Analysis seed 20261014 (no random operations); simulation seed IDs are read - `scripts/t1_specs.py`: Patched starts seeds 0..7 # Patched starts seeds 0..7 Registration/context records (combined result notes are not independent preregistration evidence): Excursion tomography: typical loops or rare journeys? (T1) — Pre-registered 2026-10-13 23:45 NZDT, before any code execution or simulation. notes/excursion-tomography-2026-10-13.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
15	experiments/2026-10-15-independent-area/results/analysis.json experiments/2026-10-15-independent-area/README.md Analysis scripts (commands in recipe): experiments/2026-10-15-independent-area/scripts/analyse.py experiments/2026-10-15-independent-area/scripts/make_notebook.py	Seed declarations: Bootstrap rules (2000, seed 20261015), keeping roles and windows together; controls (44 rules, 3 groups, 200 independent datasets, seed 20261015): SEED = 20261015 Registration/context records (combined result notes are not independent preregistration evidence): Open questions: hypotheses before measuring — Written 2026-10-10 10:30 NZDT. Data in hand: probe_summary.csv (c, kappa, ... notes/open-questions-hypotheses.md Earliest path commit: uncommitted / no tracked history
16	experiments/2026-10-14-kappa-matched-area/results/noise_floor.txt experiments/2026-10-14-kappa-matched-area/README.md Analysis scripts (commands in recipe): experiments/2026-10-14-kappa-matched-area/scripts/matched_area.py experiments/2026-10-14-kappa-matched-area/scripts/matched_pairs.py	Seed declarations: ln kappa on ln A, with rule-bootstrap 95% intervals (seed 20261014, 500 resamples). (tol 0.05 registered, 0.10 sensitivity), rule means over seeds as before. Bootstrap seed 20261016. """M1-M3 of ../README.md on the committed T1b windows. No simulation; bootstrap seed 20261014."""

Audit	Exact records and producers	Seeds, registration and Git record
	experiments/2026-10-14-kappa-matched-area/scripts/noise_floor.py	Registration/context records (combined result notes are not independent preregistration evidence): No separate registration title located in the inspected records. Earliest path commit: 741ccf3c63ec5053f31de6fb4c8ab15267158beb (2026-10-15T19:38:37+13:00)
17	notes/transport-forecast-2026-10-13.md	Seed declarations: rules 2000 times, seed 20261013; report within-rule seed sd. Full trajectory records, input SHA256 hashes, exclusions, commit and seed: 61.33–80.58% (2,000 resamples, seed 20261013). The registered 30% prediction Registration/context records (combined result notes are not independent preregistration evidence): Prospective transport forecast — Registered 2026-10-13 23:35 NZDT, before computing forecasts. Base commit notes/transport-forecast-2026-10-13.md Earliest path commit: 89d8dd6742cbf5663fdb63e5e275078c82c2d384 (2026-10-15T19:38:37+13:00)
18	experiments/2026-10-13-transport-forecast/scripts/transport_forecast.py experiments/2026-10-13-transport-forecast/results/transport_forecast.json experiments/2026-10-13-transport-forecast/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-transport-forecast/scripts/transport_forecast.py	Seed declarations: rng = np.random.default_rng(20261013) command='uv run experiments/2026-10-13-transport-forecast/scripts/transport_forecast.py', bootstrap_seed=20261013, Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: 797704a2988ea8d958df39ac96e3ece27bec0980 (2026-10-14T19:31:38+13:00)
19	notes/frontier-2026-10-13-results.md	Seed declarations: Statistic: x_k = A(2^k)/2^(2k/3), k = 24..30. Per k: CV with bootstrap SE (2000 resamples, seed 20261013). Estimator: local slope beta(k) = dlog P / dlog A between octaves k-4 and k (`edge_exponent.py` definition, P = boundary cells), rotor seeds 1..8 to 2^34. - Local area exponents a = log2(A(2^24)/A(2^22))/2 are 0.84-1.02 (median ~0.93; one run, RRRLR seed 2, is at 1.02), matching the earlier 0.92 for this ensemble. Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Calibration against the rotor (Eulerian) walker notes/calibration.md Earliest path commit: 5be7ac7f1b29e49bba217ccc4f7010e503e86c10 (2026-10-13T21:51:43+13:00)
20	notes/walker-comparisons-2026-10-13.md	Seed declarations: Run each plus random walk and rotor controls for seeds 301..308, to 2^20, fits, input hashes, commit and seed: `experiments/2026-10-13-w-width-radius-refit/results/width_radius_refit.json`. Seed 301 selected before inspecting paths; 65,536 steps, ant RLR blank start. Registration/context records (combined result notes are not independent preregistration evidence): Walker comparisons: growth and boundary width — ## Registered 2026-10-13 23:27 NZDT, before the refit

Audit	Exact records and producers	Seeds, registration and Git record
		notes/walker-comparisons-2026-10-13.md Earliest path commit: 89d8dd6742cbf5663fdb63e5e275078c82c2d384 (2026-10-15T19:38:37+13:00)
21	experiments/2026-10-13-memory-walkers/results/memory_walkers_pilot/summary.json experiments/2026-10-13-memory-walkers/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-memory-walkers/scripts/memory_walkers_pilot.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: 797704a2988ea8d958df39ac96e3ece27bec0980 (2026-10-14T19:31:38+13:00)
22	examples/layers.rs Analysis scripts (commands in recipe): examples/layers.rs	Seed declarations: run(r, Rotor { seed: seed.parse().unwrap() }, size, max_log2) Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: 563812d720a65e4baa3b01de5d65f9883000cbfa (2026-10-13T23:05:08+13:00)
23	notes/kappa-experiment-design.md	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: 563812d720a65e4baa3b01de5d65f9883000cbfa (2026-10-13T23:05:08+13:00)
24	experiments/2026-10-13-layers-colours/results/layers_stats.txt experiments/2026-10-13-layers-colours/results/layers_stats_typical.txt experiments/2026-10-13-layers-colours/results/layers_classes.txt experiments/2026-10-13-layers-colours/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-layers-colours/scripts/layers_census_c.py experiments/2026-10-13-layers-colours/scripts/layers_classes.py experiments/2026-10-13-layers-colours/scripts/layers_select.py experiments/2026-10-13-layers-colours/scripts/layers_stats.py experiments/2026-10-13-layers-colours/scripts/layers_stats_typical.py	Boundary-layer ant runs: deterministic blank boards, no random seed. Control seeds and full rule list: experiments/2026-10-13-layers-colours/results/layers_specs.txt. Registration: "What sets kappa? Experiment design", written 2026-10-13 before runs; exploratory crust classification was not predicted. Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
25	experiments/2026-10-13-arm-kappa-depletion-arms/results/armb_stats.txt experiments/2026-10-13-arm-kappa-depletion-arms/results/armc_stats.txt experiments/2026-10-13-arm-kappa-depletion-arms/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-arm-kappa-depletion-arms/scripts/armb_stats.py experiments/2026-10-13-arm-kappa-depletion-arms/scripts/armc_pairs.py experiments/2026-10-13-arm-kappa-depletion-arms/scripts/armc_stats.py	Noisy-rotor seeds: 1, 2, 3, 4, 5, 6, 7, 8. Exact input: experiments/2026-10-13-arm-kappa-depletion-arms/results/armb_specs.txt. Producer: src/bin/probe.rs; analysis: scripts/armb_stats.py. Registration: "What sets kappa? Experiment design", Arm B (2026-10-13). Actual command and deviations are in that note; the planned board and window differed from those run. Registration/context records (combined result notes are not independent preregistration evidence): What sets kappa? Experiment design (written 2026-10-13, before any run) — # What sets

Audit	Exact records and producers	Seeds, registration and Git record
		kappa? Experiment design (written 2026-10-13, before any run) notes/kappa-experiment-design.md Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
26	experiments/2026-10-13-mixing/results/mix_rules.txt experiments/2026-10-13-mixing/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-mixing/scripts/mixing_calib.py experiments/2026-10-13-mixing/scripts/mixing_stats.py	Seed declarations: rng = np.random.default_ rng(20261012) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
27	notes/mixing-random-sample.md	Seed declarations: Protocol: draw 40 rules uniformly at random (numpy default_rng seed 20261012) Registration/context records (combined result notes are not independent preregistration evidence): Mixing test on a random sample of rules — Written 2026-10-12, before running. Reason: the earlier mixing sample notes/mixing-random-sample.md Earliest path commit: 50206967c655891659ac 98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
28	experiments/2026-10-13-item4-exponent-rotor-calibrated/ scripts/item4_stats.py experiments/2026-10-13-item4-exponent-rotor-calibrated/ README.md Analysis scripts (commands in recipe): experiments/2026-10-13-item4-exponent-rotor-calibrated/ scripts/item4_select.py experiments/2026-10-13-item4-exponent-rotor-calibrated/ scripts/item4_stats.py	Seed declarations: rule=pl.col("rule").str.split(SEP).list.get(0), seed=pl.col("rule").str.split(SEP).list.get(1).cast(pl.Int64)) and over the census width exponent (low/mid/ high), numpy seed 20261012. Writes experiments/2026-10-13-item4-exponent-rotor-calibrated/results/item4_rules.txt and experiments/2026-10-13-item4-exponent-rotor-calibrated/results/item4_walkers.txt (RULE@SEED, seeds 1..24).""" Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
29	experiments/2026-10-13-shape-limit/scripts/shape_stats.py experiments/2026-10-13-shape-limit/results/shape_stats.txt experiments/2026-10-13-shape-limit/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-shape-limit/scripts/shape_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
30	experiments/2026-10-13-item21-slow-outliers/scripts/ item21_stats.py experiments/2026-10-13-item21-slow-outliers/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-item21-slow-outliers/scripts/ item21_select.py experiments/2026-10-13-item21-slow-outliers/scripts/ item21_stats.py	Seed declarations: rule_ =pl.col("rule").str.split("@").list.get(0), seed=pl.col("rule").str.split("@").list.get(1).cast(pl.Int64)) controls (0.30 < alpha_w < 0.37), numpy seed 20261013; write experiments/2026-10-13-item21-slow-outliers/results/item21_walkers.txt (RULE@SEED, seeds 1..12).""" core = s.filter((pl.col("alpha_w") > 0.30) & (pl.col("alpha_w") < 0.37)).sample(30, seed=20261013) Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8a e91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
31	notes/frontier-results.md	Seed declarations: (seeds 1..24, `RULE@SEED`), 2^30 steps, octave probe output

Audit	Exact records and producers	Seeds, registration and Git record
		- H12 (formation). Patched-start ants (5x5 random patch, seeds 1..48), 2^26 from 12 random 5x5 patches (seeds 1..12), 2^27 steps, same fit window as the census Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: 50206967c655891659ac98b2776925e9a222e773 (2026-10-13T08:14:42+13:00)
32	experiments/2026-10-13-w3/results/w3_stats.txt experiments/2026-10-13-w3/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-w3/scripts/w3_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
33	experiments/2026-10-13-w-width-radius-refit/results/w_checks.txt experiments/2026-10-13-w-width-radius-refit/results/width_radius_refit.json experiments/2026-10-13-w-width-radius-refit/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-w-width-radius-refit/scripts/w_checks.py experiments/2026-10-13-w-width-radius-refit/scripts/width_radius_refit.py	Seed declarations: SEED = 20261013 Registration/context records (combined result notes are not independent preregistration evidence): Walker comparisons: growth and boundary width — ## Registered 2026-10-13 23:27 NZDT, before the refit notes/walker-comparisons-2026-10-13.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
34	experiments/2026-10-13-a4-boundary-width/results/a4c_stats.txt experiments/2026-10-13-a4-boundary-width/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-a4-boundary-width/scripts/a4_stats.py experiments/2026-10-13-a4-boundary-width/scripts/a4c_stats.py	Seed declarations: rng = np.random.default_rng(20261013) Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
35	experiments/2026-10-14-hull-width/results/hull_stats.txt experiments/2026-10-14-hull-width/README.md Analysis scripts (commands in recipe): experiments/2026-10-14-hull-width/scripts/hull_bands.py experiments/2026-10-14-hull-width/scripts/hull_stats.py	Seed declarations: patched-start seeds 221-224 (24 runs, `RULE@SEED`), and rotor seeds 221-228 (8 runs), 8192^2 (per run); medians over runs, with between-run sd and a rule-level bootstrap (seed 20261014) for `uv run experiments/2026-10-14-hull-width/scripts/hull_bands.py` -> `results/hull_bands.txt` (no simulation; 6 ant rules x 4 seeds, 8 rotor runs; bootstrap seed 20261014). Registration/context records (combined result notes are not independent preregistration evidence): Open questions: hypotheses before measuring — Written 2026-10-10 10:30 NZDT. Data in hand: probe_summary.csv (c, kappa, ... notes/open-questions-hypotheses.md Earliest path commit: 741ccf3c63ec5053f31de6fb4c8ab15267158beb (2026-10-15T19:38:37+13:00)
36	experiments/2026-10-13-edge-statistics/scripts/edge_exponent.py experiments/2026-10-13-edge-statistics/results/edge_exponent.txt experiments/2026-10-13-edge-statistics/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-edge-statistics/scripts/edge_exponent.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)

Audit	Exact records and producers	Seeds, registration and Git record
37	experiments/2026-10-13-b5-late-highways/results/b5_stats.txt experiments/2026-10-13-b5-late-highways/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-b5-late-highways/scripts/b5_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Results ledger for `research-frontier-2026-10-13.md` — # Results ledger for `research-frontier-2026-10-13.md` notes/frontier-2026-10-13-results.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
38	examples/ribbon_trace.rs Analysis scripts (commands in recipe): examples/ribbon_trace.rs	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: 734b6822654033f16b117ca68d9c46b63564f09a (2026-10-13T23:41:31+13:00)
39	experiments/2026-10-13-g1-ribbon-breakout/results/g1_stats.txt experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_builder_*.py experiments/2026-10-13-g1-ribbon-breakout/README.md Analysis scripts (commands in recipe): experiments/2026-10-13-g1-ribbon-breakout/scripts/compare_survivors.py experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_breakout_stats.py experiments/2026-10-13-g1-ribbon-breakout/scripts/g1_stats.py	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): Research frontier (2026-10-13 PM) — # Research frontier (2026-10-13 PM) notes/research-frontier-2026-10-13-pm.md Earliest path commit: c5f061aa0f66e6f6cb8ae91163aec5e3b7897707 (2026-10-14T19:30:16+13:00)
40	examples/builder_dump.rs Analysis scripts (commands in recipe): examples/builder_dump.rs	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): No separate registration title located in the inspected records. Earliest path commit: uncommitted / no tracked history Referenced paths absent from checkout: examples/builder_dump.rs
41	notes/g1-builder-structure-2026-10-14.md	Seed schedule: see the exact recipe and per-run metadata unless specified above. Registration/context records (combined result notes are not independent preregistration evidence): notes/g1-builder-structure-2026-10-14.md — referenced record absent from this checkout Earliest path commit: uncommitted / no tracked history Referenced paths absent from checkout: notes/g1-builder-structure-2026-10-14.md
42	experiments/2026-10-15-chirality-baselines/results/comparison.json experiments/2026-10-15-chirality-baselines/results/groups.csv experiments/2026-10-15-chirality-baselines/README.md	uv run experiments/2026-10-15-chirality-baselines/scripts/compare.py uv run research-outputs/cube-root-growth-paper/chirality_baseline_numbers.py Critique-driven reanalysis; same 300 calibration and 2747 held-out rules. Paired rule bootstrap: 4000 draws, seed 20261046; fits fixed. Input hashes and base commit in comparison.json; per-rule predictions and an executed notebook accompany the results.
43	experiments/2026-10-15-kappa-tail-budget/results/confirm-v1.analysis.json	uv run experiments/2026-10-15-kappa-tail-budget/scripts/run.py --name confirm-v1

Audit	Exact records and producers	Seeds, registration and Git record
	experiments/2026-10-15-kappa-tail-budget/results/confirm-v1.decomposition.json experiments/2026-10-15-kappa-tail-budget/README.md	uv run experiments/2026-10-15-kappa-tail-budget/scripts/confirm_analysis.py uv run experiments/2026-10-15-kappa-tail-budget/scripts/confirm_decomposition.py uv run research-outputs/cube-root-growth-paper/tail_budget_numbers.py Seeds 2026101801-32 (predictor odd, outcome even offsets); bootstrap seed 2026101599, 2000 draws. Manifest sha256 19a86779... in confirm-v1.manifest.json. Registered in the README before launch; analysis and decomposition scripts written after the run (deviation entries in NOTEBOOK.md). Dumps are gitignored; a fresh clone reruns about 94 minutes.

Bibliography

[1] A. Bou-Rabee and Y. Peres, "Eulerian walkers on Z^2 have range exponent 2/3." [Online]. Available: https://arxiv.org/abs/2608.23545

[2] C. G. Langton, "Studying artificial life with cellular automata," *Physica D: Nonlinear Phenomena*, vol. 22, pp. 120–149, 1986, doi: 10.1016/0167-2789(86)90237-X.

[3] D. Gale, J. Propp, S. Sutherland, and S. Troubetzkoy, "Further travels with my ant," *The Mathematical Intelligencer*, vol. 17, no. 3, pp. 48–56, 1995, doi: 10.1007/BF03024370.

[4] L. A. Bunimovich and S. E. Troubetzkoy, "Recurrence properties of Lorentz lattice gas cellular automata," *Journal of Statistical Physics*, vol. 67, pp. 289–302, 1992, doi: 10.1007/BF01049035.

[5] A. Gajardo, A. Moreira, and E. Goles, "Complexity of Langton's ant," *Discrete Applied Mathematics*, vol. 117, pp. 41–50, 2002, doi: 10.1016/S0166-218X(00)00334-6.

[6] V. B. Priezzhev, D. Dhar, A. Dhar, and S. Krishnamurthy, "Eulerian walkers as a model of self-organized criticality," *Physical Review Letters*, vol. 77, p. 5079, 1996, doi: 10.1103/PhysRevLett.77.5079.

[7] R. Kapri and D. Dhar, "Asymptotic shape of the region visited by an Eulerian walker," *Physical Review E*, vol. 80, p. 51118, 2009, [Online]. Available: https://arxiv.org/abs/0906.5506

[8] H.-F. Meng and E. G. D. Cohen, "Growth, self-randomization, and propagation in a Lorentz lattice gas," *Physical Review E*, vol. 50, no. 4, pp. 2482–2487, 1994, doi: 10.1103/PhysRevE.50.2482.

[9] S. Mishra, S. Bhattacharya, B. Webb, and E. G. D. Cohen, "Subdiffusion, Anomalous Diffusion and Propagation of a Particle Moving in Random and Periodic Media," *Journal of Statistical Physics*, 2016, doi: 10.1007/s10955-016-1448-5.

[10] V. V. Papoyan, V. S. Poghosyan, and V. B. Priezzhev, "Rotor-Router Walk on a Semi-infinite Cylinder," *Journal of Statistical Mechanics: Theory and Experiment*, p. 73209, 2016, doi: 10.1088/1742-5468/2016/07/073209.

[11] I. Benjamini and D. B. Wilson, "Excited Random Walk." [Online]. Available: https://arxiv.org/abs/math/0302271

[12] F. Merkl and S. W. W. Rolles, "Linearly edge-reinforced random walks." [Online]. Available: https://arxiv.org/abs/math/0608220

[13] M. Kardar, G. Parisi, and Y.-C. Zhang, "Dynamic Scaling of Growing Interfaces," *Physical Review Letters*, vol. 56, pp. 889–892, 1986, doi: 10.1103/PhysRevLett.56.889.

[14] J. J. Ramasco, J. M. Lopez, and M. A. Rodriguez, "Generic Dynamic Scaling in Kinetic Roughening," *Physical Review Letters*, vol. 84, pp. 2199–2202, 2000, doi: 10.1103/PhysRevLett.84.2199.

[15] L. Florescu, L. Levine, and Y. Peres, "The range of a rotor walk," *The American Mathematical Monthly*, vol. 123, no. 7, pp. 627–642, 2016, doi: 10.4169/amer.math.monthly.123.7.627.